

C v3 Vertical Slice v1 - Stage 2 Semantic Repair Human Review

Repair validation: PASS | Human semantic audit: PENDING | Stage 3: NOT STARTED

20-point regression overview

- LI18 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- ST10 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- LI17 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- BL17 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- BL23 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- BL25 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- ST9 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- TE20 | UNRESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- HT7 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['direction_preservation_failure', 'operand_trace_incomplete', 'relation_nonoperand_semantics_loss', 'relation_semantic_hash_mismatch'] | post-repair findings=0
- LI4 | RESOLVED -> RESOLVED | pre-repair rules=['direction_preservation_failure', 'operand_trace_incomplete', 'parent_only_composite_resolution', 'relation_nonoperand_semantics_loss', 'relation_semantic_hash_mismatch'] | post-repair findings=0
- ST1 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['direction_preservation_failure', 'operand_trace_incomplete', 'relation_nonoperand_semantics_loss', 'relation_semantic_hash_mismatch'] | post-repair findings=0
- GB14 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['direction_preservation_failure', 'operand_trace_incomplete', 'quantitative_relation_semantics_loss', 'relation_nonoperand_semantics_loss', 'relation_semantic_hash_mismatch'] | post-repair findings=0
- GB23 | RESOLVED -> RESOLVED | pre-repair rules=['direction_preservation_failure', 'operand_trace_incomplete', 'quantitative_relation_semantics_loss', 'relation_nonoperand_semantics_loss', 'relation_semantic_hash_mismatch'] | post-repair findings=0
- LU6 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['direction_preservation_failure', 'operand_trace_incomplete', 'quantitative_relation_semantics_loss', 'relation_nonoperand_semantics_loss', 'relation_semantic_hash_mismatch'] | post-repair findings=0
- LI7 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['direction_preservation_failure', 'operand_trace_incomplete', 'quantitative_relation_semantics_loss', 'relation_nonoperand_semantics_loss', 'relation_semantic_hash_mismatch'] | post-repair findings=0
- GB26 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['direction_preservation_failure', 'operand_trace_incomplete', 'relation_nonoperand_semantics_loss', 'relation_semantic_hash_mismatch'] | post-repair findings=0
- ST2 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['operand_trace_incomplete', 'relation_nonoperand_semantics_loss', 'relation_semantic_hash_mismatch'] | post-repair findings=0
- BL40 | RESOLVED -> RESOLVED | pre-repair rules=['operand_trace_incomplete', 'relation_nonoperand_semantics_loss', 'relation_semantic_hash_mismatch'] | post-repair findings=0
- ST4 | RESOLVED -> UNRESOLVED | pre-repair rules=['condition_linkage_loss', 'direction_preservation_failure', 'operand_trace_incomplete', 'parent_only_composite_resolution', 'quantitative_relation_semantics_loss', 'relation_nonoperand_semantics_loss', 'relation_semantic_hash_mismatch'] | post-repair findings=0
- TE6 | RESOLVED -> RESOLVED | pre-repair rules=['direction_preservation_failure', 'operand_trace_incomplete', 'quantitative_relation_semantics_loss', 'relation_nonoperand_semantics_loss', 'relation_semantic_hash_mismatch'] | post-repair findings=0

LI18: RESOLVED -> UNRESOLVED

Defect family: composite_subfeature_collapse, modifier_dependency_loss

WHO source

location | S:LI18:location | page 51 | On the anterior aspect of the neck, at the same level as the superior border of the thyroid cartilage, between the anterior and posterior borders of the sternocleidomas-toid muscle.

Semantic decomposition

LM:S:LI18:location:115-180:0f3e9745 | source=anterior and posterior borders of the sternocleidomas-toid muscle | class=muscle | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:LI18:location:128-180:9046d75e | source=posterior borders of the sternocleidomas-toid muscle | class=muscle | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:LI18:location:57-101 | source=the superior border of the thyroid cartilage | class=cartilage | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:LI18:location:61-76 | source=superior border | class=boundary.border | parent=LM:S:LI18:location:84-101 | child=superior border | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=no approved executable construction rule for canonical subfeature 'superior_border'

LM:S:LI18:location:7-22 | source=anterior aspect | class=surface.aspect | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:LI18:location:84-101 | source=thyroid cartilage | class=cartilage | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

Composite binding view

PARENT LM:S:LI18:location:84-101 thyroid cartilage status=RESOLVED identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:LI18:location:61-76 superior border status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

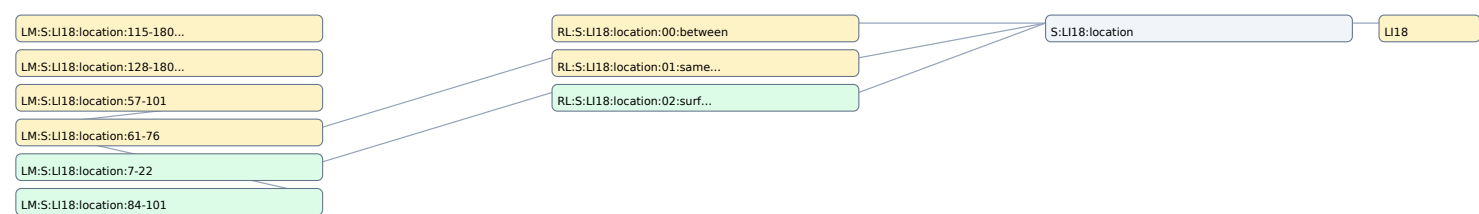
Relation execution

RL:S:LI18:location:00:between | op=between_constraint | semantic=['LM:S:LI18:location:115-180:0f3e9745', 'LM:S:LI18:location:128-180:9046d75e'] | executable=[] | dir=None | branch=None | source=S:LI18:location | semantic_hash=007fb4cbe287e79acf45359082a1b7ca5e81cdf6d7f5b11bf09224ae1cd15a41 | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:LI18:location:01:same-level | op=same_level_plane | semantic=['LM:S:LI18:location:57-101'] | executable=['LM:S:LI18:location:61-76'] | dir=None | branch=None | source=S:LI18:location | semantic_hash=46aa139fe659b5830bd200f773a7812545842e263532309e0cedaa145152f4f2 | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:LI18:location:02:surface-landmark | op=surface_membership | semantic=['LM:S:LI18:location:7-22'] | executable=['LM:S:LI18:location:7-22'] | dir=None | branch=None | source=S:LI18:location | semantic_hash=720ed0ee76cca40e34d3f46cbce78cb5935f3bf9bb51906b10b97da40845918c | status=RESOLVED | reason=None

Dependency graph



Dependency graph edge list

- RL:S:LI18:location:00:between -> S:LI18:location [relation_to_statement]
- LM:S:LI18:location:57-101 -> LM:S:LI18:location:61-76 [semantic_to_executable]
- LM:S:LI18:location:61-76 -> RL:S:LI18:location:01:same-level [executable_to_relation]
- RL:S:LI18:location:01:same-level -> S:LI18:location [relation_to_statement]
- LM:S:LI18:location:7-22 -> RL:S:LI18:location:02:surface-landmark [executable_to_relation]
- RL:S:LI18:location:02:surface-landmark -> S:LI18:location [relation_to_statement]
- LM:S:LI18:location:84-101 -> LM:S:LI18:location:61-76 [parent_to_child_subfeature]

S:LI18:location -> LI18 [statement_to_point]

BEFORE / AFTER diff

BEFORE: {"conditions": [], "relations": [{"constraint": {"argument_node_ids": ["LM:S:LI18:location:115-180:0f3e9745", "LM:S:LI18:location:128-180:9046d75e"], "branch_id": null, "op": "between_constraint", "source_statement_id": "S:LI18:location"}, {"executor": "between_constraint", "status": "RESOLVED"}], {"constraint": {"argument_node_ids": ["LM:S:LI18:location:57-101"], "branch_id": null, "op": "same_level_plane", "source_statement_id": "S:LI18:location"}, {"executor": "same_level_plane", "status": "RESOLVED"}], {"constraint": {"argument_node_ids": ["LM:S:LI18:location:7-22"], "branch_id": null, "op": "surface_membership", "source_statement_id": "S:LI18:location"}, {"executor": "surface_membership", "status": "RESOLVED"}], "status": "RESOLVED"}

AFTER: {"conditions": [], "relations": [{"executable_argument_node_ids": [], "executor": "between_constraint", "operand_binding_trace": [{"binding": "missing_executable_subfeature", "executable_operand_ids": [], "semantic_operand_id": "LM:S:LI18:location:115-180:0f3e9745"}, {"binding": "missing_executable_subfeature", "executable_operand_ids": [], "semantic_operand_id": "LM:S:LI18:location:128-180:9046d75e"}], "reason": "semantic composite has multiple child subfeatures and no unique executable operand", "semantic_argument_node_ids": ["LM:S:LI18:location:115-180:0f3e9745", "LM:S:LI18:location:128-180:9046d75e"], "semantic_fields": {"bound_measurement_ids": [], "branch_id": null, "op": "between_constraint", "quantitative_constraints": [], "relation_type": "between", "semantic_argument_node_ids": ["LM:S:LI18:location:115-180:0f3e9745", "LM:S:LI18:location:128-180:9046d75e"], "source_statement_id": "S:LI18:location", "statement_measurement_ids": []}, "semantic_fields_hash": "007fb4cbe287e79acf45359082a1b7ca5e81cdf6d7f5b11bf09224ae1cd15a41", "status": "UNRESOLVED"}], {"executable_argument_node_ids": ["LM:S:LI18:location:61-76"], "executor": "same_level_plane", "operand_binding_trace": [{"binding": "child_subfeature", "executable_operand_id": "LM:S:LI18:location:61-76", "semantic_operand_id": "LM:S:LI18:location:57-101"}], "reason": "executable operand not uniquely executable", "semantic_argument_node_ids": ["LM:S:LI18:location:57-101"], "semantic_fields": {"bound_measurement_ids": [], "branch_id": null, "op": "same_level_plane", "quantitative_constraints": [], "relation_type": "same-level", "semantic_argument_node_ids": ["LM:S:LI18:location:57-101"], "source_statement_id": "S:LI18:location", "statement_measurement_ids": []}, "semantic_fields_hash": "46aa139fe659b5830bd200f773a7812545842e263532309e0cedaa145152f4f2", "status": "UNRESOLVED"}], {"constraint": {"argument_node_ids": ["LM:S:LI18:location:7-22"], "bound_measurement_ids": [], "branch_id": null, "executable_argument_node_ids": ["LM:S:LI18:location:7-22"], "op": "surface_membership", "operand_binding_trace": [{"binding": "identity", "executable_operand_id": "LM:S:LI18:location:7-22", "semantic_operand_id": "LM:S:LI18:location:7-22"}], "quantitative_constraints": [], "relation_type": "surface-landmark", "semantic_argument_node_ids": ["LM:S:LI18:location:7-22"], "source_statement_id": "S:LI18:location", "statement_measurement_ids": []}, {"executable_argument_node_ids": ["LM:S:LI18:location:7-22"], "executor": "surface_membership", "operand_binding_trace": [{"binding": "identity", "executable_operand_id": "LM:S:LI18:location:7-22", "semantic_operand_id": "LM:S:LI18:location:7-22"}], "semantic_argument_node_ids": ["LM:S:LI18:location:7-22"], "semantic_fields": {"bound_measurement_ids": [], "branch_id": null, "op": "surface_membership", "quantitative_constraints": [], "relation_type": "surface-landmark", "semantic_argument_node_ids": ["LM:S:LI18:location:7-22"], "source_statement_id": "S:LI18:location", "statement_measurement_ids": []}, "semantic_fields_hash": "720ed0ee76cca40e34d3f46cbce78cb5935f3bf9bb51906b10b97da40845918c", "status": "RESOLVED"}], "status": "UNRESOLVED"}

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature
- [] distinct subfeatures
- [] relation operator
- [] condition state
- [] RESOLVED/UNRESOLVED
- [] ACCEPT
- [] REJECT

ST10: RESOLVED -> UNRESOLVED

Defect family: composite_subfeature_collapse

WHO source

location | S:ST10:location | page 59 | In the anterior region of the neck, at the same level as the cricoid cartilage, just anterior to the border of the sternocleido-mastoid muscle.

Semantic decomposition

LM:S:ST10:location:100-107 | source= border | class=boundary.border | parent=LM:S:ST10:location:111-142 | child= border | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=no approved executable construction rule for canonical subfeature 'border'

LM:S:ST10:location:111-142 | source=the sternocleido-mastoid muscle | class=muscle | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA13407","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:ST10:location:57-78 | source=the cricoid cartilage | class=cartilage | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA9615","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:ST10:location:7-34 | source=anterior region of the neck | class=region.body_region | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"71f7a5ab627b3c93a9aceb574aabd7ca3577688316b7de105a5effa7840dc7","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:anterior_neck","subfeature_type":null} | reason=None

LM:S:ST10:location:96-142 | source= the border of the sternocleido-mastoid muscle | class=muscle | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

Composite binding view

PARENT LM:S:ST10:location:111-142 the sternocleido-mastoid muscle status=RESOLVED
identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA13407","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:ST10:location:100-107 border status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

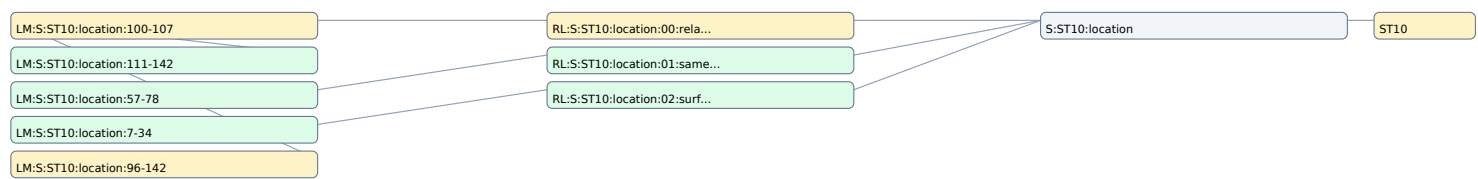
Relation execution

RL:S:ST10:location:00:relative-to | op=directional_relation | semantic=['LM:S:ST10:location:96-142'] | executable=['LM:S:ST10:location:100-107'] | dir=anterior | branch=None | source=S:ST10:location | semantic_hash=57d997e42d50177c6fb0b9bb08fdc49a0516b619e22ae62807cfc1074b3568ed | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:ST10:location:01:same-level | op=same_level_plane | semantic=['LM:S:ST10:location:57-78'] | executable=['LM:S:ST10:location:57-78'] | dir=None | branch=None | source=S:ST10:location | semantic_hash=7dd8a4b0fb5aedf2d6322e77f02276a53defc9579436788990d35b7de3cb05fd | status=RESOLVED | reason=None

RL:S:ST10:location:02:surface-landmark | op=surface_membership | semantic=['LM:S:ST10:location:7-34'] | executable=['LM:S:ST10:location:7-34'] | dir=None | branch=None | source=S:ST10:location | semantic_hash=9f0cd6f1c64582850c517507079fee82f61fad0dc9ef43e1f050c19da3b2c38 | status=RESOLVED | reason=None

Dependency graph



Dependency graph edge list

- LM:S:ST10:location:96-142 -> LM:S:ST10:location:100-107 [semantic_to_executable]
- LM:S:ST10:location:100-107 -> RL:S:ST10:location:00:relative-to [executable_to_relation]
- RL:S:ST10:location:00:relative-to -> S:ST10:location [relation_to_statement]
- LM:S:ST10:location:57-78 -> RL:S:ST10:location:01:same-level [executable_to_relation]
- RL:S:ST10:location:01:same-level -> S:ST10:location [relation_to_statement]
- LM:S:ST10:location:7-34 -> RL:S:ST10:location:02:surface-landmark [executable_to_relation]
- RL:S:ST10:location:02:surface-landmark -> S:ST10:location [relation_to_statement]
- LM:S:ST10:location:111-142 -> LM:S:ST10:location:100-107 [parent_to_child_subfeature]
- S:ST10:location -> ST10 [statement_to_point]

BEFORE / AFTER diff

BEFORE: {"conditions":[],"relations":[{"constraint":{"argument_node_ids":["LM:S:ST10:location:96-142"],"branch_id":null,"direction":"anterior","op":"directional_relation","source_statement_id":"S:ST10:location"},"executor":"directional_relation","status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:ST10:location:57-78"],"branch_id":null,"op":"same_level_plane","source_statement_id":"S:ST10:location"},"executor":"same_level_plane","status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:ST10:location:7-34"],"branch_id":null,"op":"surface_membership","source_statement_id":"S:ST10:location"},"executor":"surface_membership","status":"RESOLVED"}],"status":"RESOLVED"}

AFTER: {"conditions":[],"relations":[{"executable_argument_node_ids":["LM:S:ST10:location:100-107"],"executor":"directional_relation","operand_binding_trace":{"binding":"child_subfeature","executable_operand_id":"LM:S:ST10:location:100-107","semantic_operand_id":"LM:S:ST10:location:96-142"},"reason":"executable operand not uniquely executable","semantic_argument_node_ids":["LM:S:ST10:location:96-142"],"semantic_fields":{"bound_measurement_ids":[],"branch_id":null,"direction":"anterior","op":"directional_relation","quantitative_constraints":[],"relation_type":"relative-to","semantic_argument_node_ids":["LM:S:ST10:location:96-142"],"source_statement_id":"S:ST10:location","statement_measurement_ids":[],"semantic_fields_hash":"57d997e42d50177c6fb0b9bb08fdc49a0516b619e22ae62807cfc1074b3568ed"},"status":"UNRESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:ST10:location:57-78"],"bound_measurement_ids":[],"branch_id":null,"executable_argument_node_ids":["LM:S:ST10:location:57-78"],"op":"same_level_plane","operand_binding_trace":{"binding":"identity","executable_operand_id":"LM:S:ST10:location:57-78","semantic_operand_id":"LM:S:ST10:location:57-78"},"quantitative_constraints":[],"relation_type":"same-level","semantic_argument_node_ids":["LM:S:ST10:location:57-78"],"source_statement_id":"S:ST10:location","statement_measurement_ids":[],"executable_argument_node_ids":["LM:S:ST10:location:57-78"],"executor":"same_level_plane","operand_binding_trace":{"binding":"identity","executable_operand_id":"LM:S:ST10:location:57-78","semantic_operand_id":"LM:S:ST10:location:57-78"},"semantic_argument_node_ids":["LM:S:ST10:location:57-78"],"semantic_fields":{"bound_measurement_ids":[],"branch_id":null,"op":"same_level_plane","quantitative_constraints":[],"relation_type":"same-level","semantic_argument_node_ids":["LM:S:ST10:location:57-78"],"source_statement_id":"S:ST10:location","statement_measurement_ids":[],"semantic_fields_hash":"7dd8a4b0fb5aedf2d6322e77f02276a53defc9579436788990d35b7de3cb05fd"},"status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:ST10:location:7-34"],"bound_measurement_ids":[],"branch_id":null,"executable_argument_node_ids":["LM:S:ST10:location:7-34"],"op":"surface_membership","operand_binding_trace":{"binding":"identity","executable_operand_id":"LM:S:ST10:location:7-34","semantic_operand_id":"LM:S:ST10:location:7-34"},"quantitative_constraints":[],"relation_type":"surface-landmark","semantic_argument_node_ids":["LM:S:ST10:location:7-34"],"source_statement_id":"S:ST10:location","statement_measurement_ids":[],"executable_argument_node_ids":["LM:S:ST10:location:7-34"],"executor":"surface_membership","operand_binding_trace":{"binding":"identity","executable_operand_id":"LM:S:ST10:location:7-34","semantic_operand_id":"LM:S:ST10:location:7-34"},"semantic_argument_node_ids":["LM:S:ST10:location:7-34"],"semantic_fields":{"bound_measurement_ids":[],"branch_id":null,"op":"surface_membership","quantitative_constraints":[],"relation_type":"surface-landmark","semantic_argument_node_ids":["LM:S:ST10:location:7-34"],"source_statement_id":"S:ST10:location","statement_measurement_ids":[],"semantic_fields_hash":"9f0cd6f1c64582850c517507079fee82f61fadcd0c9ef43e1f050c19da3b2c38"},"status":"UNRESOLVED"}],"status":"UNRESOLVED"}

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature
- [] distinct subfeatures
- [] relation operator
- [] condition state
- [] RESOLVED/UNRESOLVED
- [] ACCEPT
- [] REJECT

LI17: RESOLVED -> UNRESOLVED

Defect family: composite_subfeature_collapse

WHO source

location | S:LI17:location | page 51 | On the anterior aspect of the neck, at the same level as the cricoid cartilage, just posterior to the border of the sternocleido-mastoid muscle.

note | S:LI17:note:1 | page 51 | Note: Directly inferior to LI18, at the same level as St10.

Semantic decomposition

LM:S:LI17:location:101-108 | source= border | class=boundary.border | parent=LM:S:LI17:location:112-143 | child= border | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=no approved executable construction rule for canonical subfeature 'border'

LM:S:LI17:location:112-143 | source=the sternocleido-mastoid muscle | class=muscle | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA13407","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:LI17:location:57-78 | source=the cricoid cartilage | class=cartilage | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA9615","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:LI17:location:7-22 | source=anterior aspect | class=surface.aspect | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"71f7a5ab627b3c93a9aceb574aabdb27ca3577688316b7de105a5effa7840dc7","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:anterior_neck","subfeature_type":null} | reason=None

LM:S:LI17:location:97-143 | source= the border of the sternocleido-mastoid muscle | class=muscle | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:LI17:note:1:26-31 | source= LI18 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"LI18","registry_id":null,"subfeature_type":null} | reason=None

LM:S:LI17:note:1:54-58 | source=St10 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"ST10","registry_id":null,"subfeature_type":null} | reason=None

Composite binding view

PARENT LM:S:LI17:location:112-143 the sternocleido-mastoid muscle status=RESOLVED
identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA13407","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:LI17:location:101-108 border status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

Relation execution

RL:S:LI17:location:00:relative-to | op=directional_relation | semantic=['LM:S:LI17:location:97-143'] | executable=['LM:S:LI17:location:101-108'] | dir=posterior | branch=None | source=S:LI17:location | semantic_hash=a6073df0620ba4ad883ee3445f1fd5677e58914335d02d372defad9ece4cb61 | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:LI17:location:01:same-level | op=same_level_plane | semantic=['LM:S:LI17:location:57-78'] | executable=['LM:S:LI17:location:57-78'] | dir=None | branch=None | source=S:LI17:location | semantic_hash=b071b7ba0b4915588897042790da7d439272e1c8a70950d4e48e3c826c3bc593 | status=RESOLVED | reason=None

RL:S:LI17:location:02:surface-landmark | op=surface_membership | semantic=['LM:S:LI17:location:7-22'] | executable=['LM:S:LI17:location:7-22'] | dir=None | branch=None | source=S:LI17:location | semantic_hash=a671dfe031de6e2c84ed0ebcc2f05d5357be86ed60ffa06eb672a50e02e7690 | status=RESOLVED | reason=None

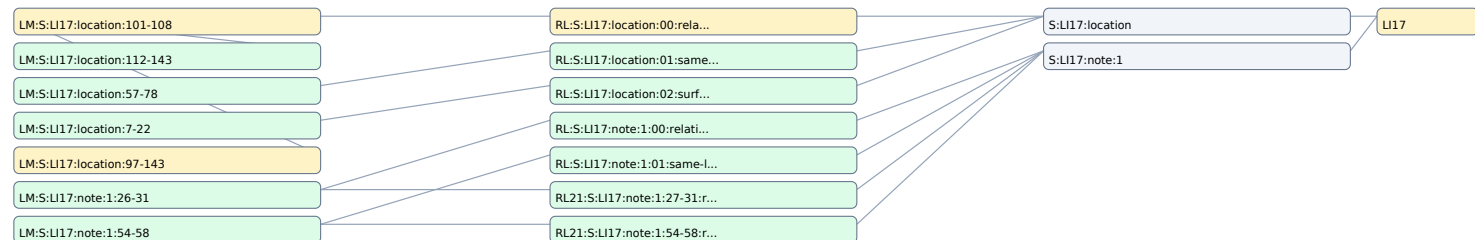
RL:S:LI17:note:1:00:relative-to | op=directional_relation | semantic=['LM:S:LI17:note:1:26-31'] | executable=['LM:S:LI17:note:1:26-31'] | dir=inferior | branch=None | source=S:LI17:note:1 | semantic_hash=da320b4d876c470c3951d9258834233f8fb76b408f3edd25f6090135fe56aa86 | status=RESOLVED | reason=None

RL:S:LI17:note:1:01:same-level | op=same_level_plane | semantic=['LM:S:LI17:note:1:54-58'] | executable=['LM:S:LI17:note:1:54-58'] | dir=None | branch=None | source=S:LI17:note:1 | semantic_hash=1f4ce7d866dff8bf5ffda4860287b78cb3a87c0ada43369442f2e98adc3d9e0c | status=RESOLVED | reason=None

RL21:S:LI17:note:1:27-31:reference-acupoint:9ab9a5cce4 | op=reference_dependency | semantic=['LM:S:LI17:note:1:26-31'] | executable=['LM:S:LI17:note:1:26-31'] | dir=None | branch=None | source=S:LI17:note:1 | semantic_hash=b838a83d82a0f14304133637a2af7d0f762d29be0a742244eb0587cf7733e0e2 | status=RESOLVED | reason=None

RL21:S:LI17:note:1:54-58:reference-acupoint:87e9101b1e | op=reference_dependency | semantic=['LM:S:LI17:note:1:54-58'] | executable=['LM:S:LI17:note:1:54-58'] | dir=None | branch=None | source=S:LI17:note:1 | semantic_hash=a7af2a878b89201ee126e162c1fea574c053927b23fd883dbe5cd07852afadfc | status=RESOLVED | reason=None

Dependency graph



Dependency graph edge list

LM:S:LI17:location:97-143 -> LM:S:LI17:location:101-108 [semantic_to_executable]

LM:S:LI17:location:101-108 -> RL:S:LI17:location:00:relative-to [executable_to_relation]

RL:S:LI17:location:00:relative-to -> S:LI17:location [relation_to_statement]

LM:S:LI17:location:57-78 -> RL:S:LI17:location:01:same-level [executable_to_relation]

RL:S:LI17:location:01:same-level -> S:LI17:location [relation_to_statement]

LM:S:LI17:location:7-22 -> RL:S:LI17:location:02:surface-landmark [executable_to_relation]

RL:S:LI17:location:02:surface-landmark -> S:LI17:location [relation_to_statement]

LM:S:LI17:note:1:26-31 -> RL:S:LI17:note:1:00:relative-to [executable_to_relation]

RL:S:LI17:note:1:00:relative-to -> S:LI17:note:1 [relation_to_statement]

LM:S:LI17:note:1:54-58 -> RL:S:LI17:note:1:01:same-level [executable_to_relation]

RL:S:LI17:note:1:01:same-level -> S:LI17:note:1 [relation_to_statement]

LM:S:LI17:note:1:26-31 -> RL21:S:LI17:note:1:27-31:reference-acupoint:9ab9a5cce4 [executable_to_relation]

RL21:S:LI17:note:1:27-31:reference-acupoint:9ab9a5cce4 -> S:LI17:note:1 [relation_to_statement]

LM:S:LI17:note:1:54-58 -> RL21:S:LI17:note:1:54-58:reference-acupoint:87e9101b1e [executable_to_relation]

RL21:S:LI17:note:1:54-58:reference-acupoint:87e9101b1e -> S:LI17:note:1 [relation_to_statement]

LM:S:LI17:location:112-143 -> LM:S:LI17:location:101-108 [parent_to_child_subfeature]

S:LI17:location -> LI17 [statement_to_point]

S:LI17:note:1 -> LI17 [statement_to_point]

BEFORE / AFTER diff

BEFORE: {"conditions":[], "relations":[{"constraint":{"argument_node_ids":["LM:S:LI17:location:97-143"], "branch_id":null, "direction":"posterior", "op":"directional_relation", "source_statement_id":"S:LI17:location"}, {"executor":"directional_relation", "status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:LI17:location:57-78"], "branch_id":null, "op":"same_level_plane", "source_statement_id":"S:LI17:location"}, {"executor":"same_level_plane", "status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:LI17:location:7-22"], "branch_id":null, "op":"surface_membership", "source_statement_id":"S:LI17:location"}, {"executor":"surface_membership", "status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:LI17:note:1:26-31"], "branch_id":null, "direction":"inferior", "op":"directional_relation", "source_statement_id":"S:LI17:note:1"}, {"executor":"directional_relation", "status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:LI17:note:1:54-58"], "branch_id":null, "op":"same_level_plane", "source_statement_id":"S:LI17:note:1"}, {"executor":"same_level_plane", "status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:LI17:note:1:26-31"], "branch_id":null, "op":"reference_dependency", "source_statement_id":"S:LI17:note:1"}, {"executor":"reference_dependency", "status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:LI17:note:1:54-58"], "branch_id":null, "op":"reference_dependency", "source_statement_id":"S:LI17:note:1"}, {"executor":"reference_dependency", "status":"RESOLVED"}]}]}

AFTER: {"conditions":[], "relations":[{"executable_argument_node_ids":["LM:S:LI17:location:101-108"], "executor":"directional_relation", "operand_binding_trace":[{"binding":"child_subfeature", "executable_operand_id":"LM:S:LI17:location:101-108", "semantic_operand_id":"LM:S:LI17:location:97-143"}], "reason":"executable operand not uniquely executable", "semantic_argument_node_ids":["LM:S:LI17:location:97-143"], "semantic_fields":{"bound_measurement_ids":[], "branch_id":null, "direction":"posterior", "op":"directional_relation", "quantitative_constraints":[], "relation_type":"relative-to", "semantic_argument_node_ids":["LM:S:LI17:location:97-143"], "source_statement_id":"S:LI17:location", "statement_measurement_ids":[], "semantic_fields_hash":"a6073df0620ba4ad883ee3445f1fd5677e58914335d02d372defad9ece4cb61", "status":"UNRESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:LI17:location:57-78"], "bound_measurement_ids":[], "branch_id":null, "executable_argument_node_ids":["LM:S:LI17:location:57-78"], "op":"same_level_plane", "operand_binding_trace":[{"binding":"identity", "executable_operand_id":"LM:S:LI17:location:57-78"}, {"semantic_operand_id":"LM:S:LI17:location:57-78", "semantic_operand_id":"LM:S:LI17:location:57-78"}], "quantitative_constraints":[], "relation_type":"same-level", "semantic_argument_node_ids":["LM:S:LI17:location:57-78"], "source_statement_id":"S:LI17:location", "statement_measurement_ids":[], "executable_argument_node_ids":["LM:S:LI17:location:57-78"], "executor":"same_level_plane", "operand_binding_trace":[{"binding":"identity", "executable_operand_id":"LM:S:LI17:location:57-78"}, {"semantic_operand_id":"LM:S:LI17:location:57-78"}], "semantic_argument_node_ids":["LM:S:LI17:location:57-78"], "semantic_fields":{"bound_measurement_ids":[], "branch_id":null, "op":"same_level_plane", "quantitative_constraints":[], "relation_type":"same-level", "semantic_argument_node_ids":["LM:S:LI17:location:57-78"], "source_statement_id":"S:LI17:location", "statement_measurement_ids":[], "semantic_fields_hash":"b071b7ba0b4915588897042790da7d439272e1c8a70950d4e48e3c826c3bc593", "status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:LI17:location:7-22"], "bound_measurement_ids":[], "branch_id":null, "executable_argument_node_ids":["LM:S:LI17:location:7-22"], "op":"surface_membership", "operand_binding_trace":[{"binding":"identity", "executable_operand_id":"LM:S:LI17:location:7-22"}, {"semantic_operand_id":"LM:S:LI17:location:7-22", "semantic_operand_id":"LM:S:LI17:location:7-22"}], "quantitative_constraints":[], "relation_type":"surface-landmark", "semantic_argument_node_ids":["LM:S:LI17:location:7-22"], "source_statement_id":"S:LI17:location", "statement_measurement_ids":[], "executable_argument_node_ids":["LM:S:LI17:location:7-22"], "executor":"surface_membership", "operand_binding_trace":[{"binding":"identity", "executable_operand_id":"LM:S:LI17:location:7-22"}, {"semantic_operand_id":"LM:S:LI17:location:7-22", "semantic_operand_id":"LM:S:LI17:location:7-22"}], "semantic_argument_node_ids":["LM:S:LI17:location:7-22"], "semantic_fields":{"bound_measurement_ids":[], "branch_id":null, "op":"surface_membership", "quantitative_constraints":[], "relation_type":"surface-landmark", "semantic_argument_node_ids":["LM:S:LI17:location:7-22"], "source_statement_id":"S:LI17:location", "statement_measurement_ids":[], "semantic_fields_hash":"a671dfe031de62c84ed0ebcc2f05d5357be86ed60ffa06eb672a50e02e7690", "status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:LI17:note:1:26-31"], "bound_measurement_ids":[], "branch_id":null, "direction":"inferior", "executable_argument_node_ids":["LM:S:LI17:note:1:26-31"], "op":"directional_relation", "operand_binding_trace":[{"binding":"identity", "executable_operand_id":"LM:S:LI17:note:1:26-31"}, {"semantic_operand_id":"LM:S:LI17:note:1:26-31"}], "quantitative_constraints":[], "relation_type":"relative-to", "semantic_argument_node_ids":["LM:S:LI17:note:1:26-31"], "source_statement_id":"S:LI17:note:1", "statement_measurement_ids":[], "executable_argument_node_ids":["LM:S:LI17:note:1:26-31"], "executor":"directional_relation", "operand_binding_trace":[{"binding":"identity", "executable_operand_id":"LM:S:LI17:note:1:26-31"}, {"semantic_operand_id":"LM:S:LI17:note:1:26-31", "semantic_operand_id":"LM:S:LI17:note:1:26-31"}], "semantic_argument_node_ids":["LM:S:LI17:note:1:26-31"], "semantic_fields":{"bound_measurement_ids":[], "branch_id":null, "direction":"inferior", "op":"directional_relation", "quantitative_constraints":[], "relation_type":"relative-to", "semantic_argument_node_ids":["LM:S:LI17:note:1:26-31"], "source_statement_id":"S:LI17:note:1", "statement_measurement_ids":[], "semantic_fields_hash":"da320b4d876c470c3951d9258834233f8fb76b408f3edd25f6090135fe56aa86", "status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:LI17:note:1:54-58"], "bound_measurement_ids":[], "branch_id":null, "executable_argument_node_ids":["LM:S:LI17:note:1:54-58"], "op":"same_level_plane", "operand_binding_trace":[{"binding":"identity", "executable_operand_id":"LM:S:LI17:note:1:54-58", "semantic_operand_id":"LM:S:LI17:note:1:54-58"}], "quantitative_constraints":[], "relation_type":"sa

me-level", "semantic_argument_node_ids":["LM:S:LI17:note:1:54-58"], "source_statement_id":"S:LI17:note:1", "statement_measurement_ids":[]}, "executable_argument_node_ids":["LM:S:LI17:note:1:54-58"], "executor":"same_level_plane", "operand_binding_trace":[{"binding":"identity", "executable_operand_id":"LM:S:LI17:note:1:54-58", "semantic_operand_id":"LM:S:LI17:note:1:54-58"}], "semantic_argument_node_ids":["LM:S:LI17:note:1:54-58"], "semantic_fields":{"bound_measurement_ids":[], "branch_id":null, "op":"same_level_plane", "quantitative_constraints":[], "relation_type":"same-level", "semantic_argument_node_ids":["LM:S:LI17:note:1:54-58"], "source_statement_id":"S:LI17:note:1", "statement_measurement_ids":[]}, "semantic_fields_hash":"1f4ce7d866dff8bf5ffda4860287b78cb3a87c0ada43369442f2e98adc3d9e0c", "status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:LI17:note:1:26-31"], "bound_measurement_ids":[], "branch_id":null, "executable_argument_node_ids":["LM:S:LI17:note:1:26-31"], "op":"reference_dependency", "operand_binding_trace":[{"binding":"identity", "executable_operand_id":"LM:S:LI17:note:1:26-31", "semantic_operand_id":"LM:S:LI17:note:1:26-31"}], "quantitative_constraints":[], "relation_type":"reference-acupoint", "semantic_argument_node_ids":["LM:S:LI17:note:1:26-31"], "source_statement_id":"S:LI17:note:1", "statement_measurement_ids":[]}, "executable_argument_node_ids":["LM:S:LI17:note:1:26-31"], "executor":"reference_dependency", "operand_binding_trace":[{"binding":"identity", "executable_operand_id":"LM:S:LI17:note:1:26-31", "semantic_operand_id":"LM:S:LI17:note:1:26-31"}], "semantic_argument_node_ids":["LM:S:LI17:note:1:26-31"], "semantic_fields":{"bound_measurement_ids":[], "branch_id":null, "op":"reference_dependency", "quantitative_constraints":[], "relation_type":"reference-acupoint", "semantic_argument_node_ids":["LM:S:LI17:note:1:26-31"], "source_statement_id":"S:LI17:note:1", "statement_measurement_ids":[]}, "semantic_fields_hash":"b838a83d82a0f14304133637a2af7d0f762d29be0a742244eb0587cf7733e0e2", "status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:LI17:note:1:54-58"], "bound_measurement_ids":[], "branch_id":null, "executable_argument_node_ids":["LM:S:LI17:note:1:54-58"], "op":"reference_dependency", "operand_binding_trace":[{"binding":"identity", "executable_operand_id":"LM:S:LI17:note:1:54-58", "semantic_operand_id":"LM:S:LI17:note:1:54-58"}], "quantitative_constraints":[], "relation_type":"reference-acupoint", "semantic_argument_node_ids":["LM:S:LI17:note:1:54-58"], "source_statement_id":"S:LI17:note:1", "statement_measurement_ids":[]}, "executable_argument_node_ids":["LM:S:LI17:note:1:54-58"], "executor":"reference_dependency", "operand_binding_trace":[{"binding":"identity", "executable_operand_id":"LM:S:LI17:note:1:54-58", "semantic_operand_id":"LM:S:LI17:note:1:54-58"}], "semantic_argument_node_ids":["LM:S:LI17:note:1:54-58"], "semantic_fields":{"bound_measurement_ids":[], "branch_id":null, "op":"reference_dependency", "quantitative_constraints":[], "relation_type":"reference-acupoint", "semantic_argument_node_ids":["LM:S:LI17:note:1:54-58"], "source_statement_id":"S:LI17:note:1", "statement_measurement_ids":[]}, "semantic_fields_hash":"a7af2a878b89201ee126e162c1fea574c053927b23fd883dbe5cd07852afadfc", "status":"UNRESOLVED"}]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature
- [] distinct subfeatures
- [] relation operator
- [] condition state
- [] RESOLVED/UNRESOLVED
- [] ACCEPT
- [] REJECT

BL17: RESOLVED -> UNRESOLVED

Defect family: modifier_dependency_loss

WHO source

location | S:BL17:location | page 117 | In the upper back region, at the same level as the inferior border of the spinous process of the seventh thoracic vertebra (T7), 1.5 B-cun lateral to the posterior median line.

note | S:BL17:note:1 | page 117 | Note: The inferior angle of the scapula is at the same level as the spinous process of the seventh thoracic vertebra.

Semantic decomposition

LM:S:BL17:location:149-175 | source= the posterior median line | class=line.anatomical_line | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"62a4312f07e04de5ebb4791906ed7098bb7c85bea907953c59c8c92bec7d9f4d","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:posterior_median_line","subfeature_type":null} | reason=None

LM:S:BL17:location:47-127 | source=the inferior border of the spinous process of the seventh thoracic vertebra (T7) | class=bone.bone | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:BL17:location:51-66 | source=inferior border | class=boundary.border | parent=LM:S:BL17:location:70-127 | child=inferior border | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=no approved executable construction rule for canonical subfeature 'inferior_border'

LM:S:BL17:location:7-24 | source=upper back region | class=region.body_region | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"86974891de98cccd457e268703bc6a672fce4777ca5d0776994671dd7f004323","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:upper_back","subfeature_type":null} | reason=None

LM:S:BL17:location:70-127 | source=the spinous process of the seventh thoracic vertebra (T7) | class=bone.bone | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:BL17:note:1:64-116 | source=the spinous process of the seventh thoracic vertebra | class=bone.bone | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:BL17:note:1:75-83 | source= process | class=bone.process_or_prominence | parent=LM:S:BL17:note:1:87-116 | child= process | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=no approved executable construction rule for canonical subfeature 'process'

LM:S:BL17:note:1:87-116 | source=the seventh thoracic vertebra | class=bone.bone | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA9968","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

Composite binding view

PARENT LM:S:BL17:location:70-127 the spinous process of the seventh thoracic vertebra (T7) status=UNRESOLVED identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:BL17:location:51-66 inferior border status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

PARENT LM:S:BL17:note:1:87-116 the seventh thoracic vertebra status=RESOLVED identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA9968","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:BL17:note:1:75-83 process status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

Relation execution

RL:S:BL17:location:00:relative-to | op=directional_relation | semantic=['LM:S:BL17:location:149-175'] | executable=['LM:S:BL17:location:149-175'] | dir=lateral | branch=None | source=S:BL17:location | semantic_hash=eb90431572e54d85b813e5238def07b5e843da57b98f049974677c15a362516e | status=RESOLVED | reason=None

RL:S:BL17:location:01:same-level | op=same_level_plane | semantic=['LM:S:BL17:location:47-127'] | executable=['LM:S:BL17:location:51-66'] | dir=None | branch=None | source=S:BL17:location | semantic_hash=07292fe70ec5db0e176595e03c0b5e5d048461aa19211c1d800f4f9e2b858b9c | status=UNRESOLVED | reason=executable operand not uniquely executable

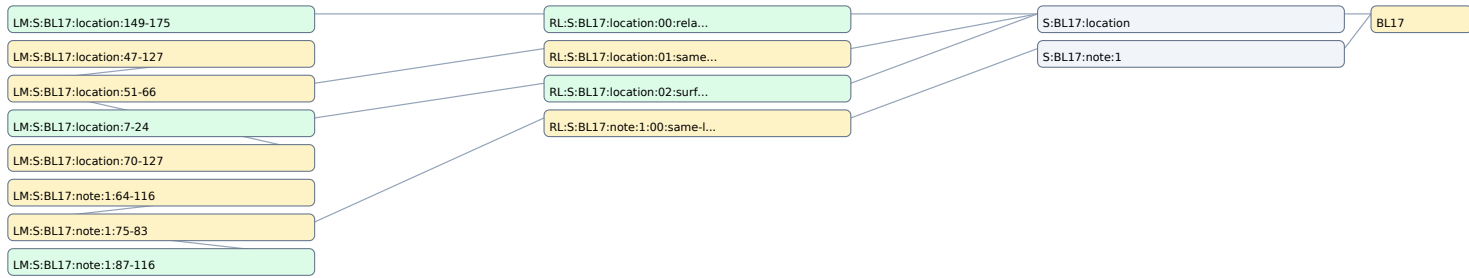
RL:S:BL17:location:02:surface-landmark | op=surface_membership | semantic=['LM:S:BL17:location:7-24'] | executable=['LM:S:BL17:location:7-24'] | dir=None | branch=None | source=S:BL17:location | semantic_hash=cf359676f8008c132b4047d2f520020bcb7c89a8e7af3f37f2db1f6571444149 | status=RESOLVED | reason=None

RL:S:BL17:note:1:00:same-level | op=same_level_plane | semantic=['LM:S:BL17:note:1:64-116'] | executable=['LM:S:BL17:note:1:75-83'] | dir=None | branch=None | source=S:BL17:note:1 | semantic_hash=31a6d961162176ebafd9aa3c27d47efedeec7a2953c1f917973bc0d9b90b2f85 | status=UNRESOLVED | reason=executable operand not uniquely executable

Quantitative measurement semantics

{ "anchor_landmark_id": "LM:S:BL17:location:149-175", "direction": "lateral", "executor": "approved_calibration", "measurement_id": "PM:S:BL17:location:00", "reason": null, "source_statement_id": "S:BL17:location", "source_unit": "B-cun", "status": "RESOLVED", "unit": "B-cun", "value": 1.5, "who_pdf_page": 117 }

Dependency graph



Dependency graph edge list

LM:S:BL17:location:149-175 -> RL:S:BL17:location:00:relative-to [executable_to_relation]

PM:S:BL17:location:00 -> RL:S:BL17:location:00:relative-to [measurement_to_relation]

RL:S:BL17:location:00:relative-to -> S:BL17:location [relation_to_statement]

LM:S:BL17:location:47-127 -> LM:S:BL17:location:51-66 [semantic_to_executable]

LM:S:BL17:location:51-66 -> RL:S:BL17:location:01:same-level [executable_to_relation]

RL:S:BL17:location:01:same-level -> S:BL17:location [relation_to_statement]

LM:S:BL17:location:7-24 -> RL:S:BL17:location:02:surface-landmark [executable_to_relation]

RL:S:BL17:location:02:surface-landmark -> S:BL17:location [relation_to_statement]

LM:S:BL17:note:1:64-116 -> LM:S:BL17:note:1:75-83 [semantic_to_executable]

LM:S:BL17:note:1:75-83 -> RL:S:BL17:note:1:00:same-level [executable_to_relation]

RL:S:BL17:note:1:00:same-level -> S:BL17:note:1 [relation_to_statement]

LM:S:BL17:location:70-127 -> LM:S:BL17:location:51-66 [parent_to_child_subfeature]

LM:S:BL17:note:1:87-116 -> LM:S:BL17:note:1:75-83 [parent_to_child_subfeature]

LM:S:BL17:location:149-175 -> PM:S:BL17:location:00 [anchor_to_measurement]

PM:S:BL17:location:00 -> S:BL17:location [measurement_to_statement]

S:BL17:location -> BL17 [statement_to_point]

S:BL17:note:1 -> BL17 [statement_to_point]

BEFORE / AFTER diff

BEFORE: {"conditions":[],"relations":[{"constraint":{"argument_node_ids":["LM:S:BL17:location:149-175"],"branch_id":null,"direction":"lateral","op":"directional_relation","source_statement_id":"S:BL17:location"},"executor":"directional_relation","status":"RESOLVED"},{"constraint":{"argument_node_ids":["LM:S:BL17:location:47-127"],"branch_id":null,"op":"same_level_plane","source_statement_id":"S:BL17:location"},"executor":"same_level_plane","status":"RESOLVED"},{"constraint":{"argument_node_ids":["LM:S:BL17:location:7-24"],"branch_id":null,"op":"surface_membership","source_statement_id":"S:BL17:location"},"executor":"surface_membership","status":"RESOLVED"},{"constraint":{"argument_node_ids":["LM:S:BL17:note:1:64-116"],"branch_id":null,"op":"same_level_plane","source_statement_id":"S:BL17:note:1"},"executor":"same_level_plane","status":"RESOLVED"}],"status":"RESOLVED"}

AFTER: {"conditions":[],"relations":[{"constraint":{"argument_node_ids":["LM:S:BL17:location:149-175"],"bound_measurement_ids":["PM:S:BL17:location:00"],"branch_id":null,"direction":"lateral","executable_argument_node_ids":["LM:S:BL17:location:149-175"],"op":"directional_relation","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM:S:BL17:location:149-175","semantic_operand_id":"LM:S:BL17:location:149-175"}],"quantitative_constraints":[{"anchor_landmark_id":"LM:S:BL17:location:149-175","direction":"lateral","kind":"proportional_offset","source_statement_id":"S:BL17:location","source_unit":"B-cun","unit":"B-cun","value":1.5,"who_pdf_page":117}],"relation_type":"relative-to","semantic_argument_node_ids":["LM:S:BL17:location:149-175"],"source_statement_id":"S:BL17:location","statement_measurement_ids":["PM:S:BL17:location:00"]},"executable_argument_node_ids":["LM:S:BL17:location:149-175"],"executor":"directional_relation","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM:S:BL17:location:149-175","semantic_operand_id":"LM:S:BL17:location:149-175"}],"semantic_argument_node_ids":["LM:S:BL17:location:149-175"],"semantic_fields":{"bound_measurement_ids":["PM:S:BL17:location:00"],"branch_id":null,"direction":"lateral","op":"directional_relation","quantitative_constraints":[{"anchor_landmark_id":"LM:S:BL17:location:149-175","direction":"lateral","kind":"proportional_offset","source_statement_id":"S:BL17:location","source_unit":"B-cun","unit":"B-cun","value":1.5,"who_pdf_page":117}],"relation_type":"relative-to","semantic_argument_node_ids":["LM:S:BL17:location:149-175"],"source_statement_id":"S:BL17:location","statement_measurement_ids":["PM:S:BL17:location:00"]},"semantic_fields_hash":"eb90431572e54d85b813e5238def07b5e843da57b98f049974677c15a362516e","status":"RESOLVED"},{"constraint":{"argument_node_ids":["LM:S:BL17:location:51-66"],"branch_id":null,"op":"same_level_plane","source_statement_id":"LM:S:BL17:location:47-127"},"executable_argument_node_ids":["LM:S:BL17:location:51-66"],"executor":"same_level_plane","operand_binding_trace":[{"binding":"child_subfeature","executable_operand_id":"LM:S:BL17:location:51-66","semantic_operand_id":"LM:S:BL17:location:47-127"}],"reason":"executable operand not uniquely executable","semantic_argument_node_ids":["LM:S:BL17:location:47-127"],"semantic_fields":{"bound_measurement_ids":["LM:S:BL17:location:47-127"],"branch_id":null,"op":"same_level_plane","quantitative_constraints":[{"relation_type":"same-level","semantic_argument_node_ids":["LM:S:BL17:location:47-127"],"source_statement_id":"S:BL17:location","statement_measurement_ids":["PM:S:BL17:location:00"]},"semantic_fields_hash":"07292fe70ec5db0e176595e03c0b5e5d048461aa19211c1d800f4f9e2b858b9c","status":"UNRESOLVED"},{"constraint":{"argument_node_ids":["LM:S:BL17:location:7-24"],"bound_measurement_ids":["LM:S:BL17:location:7-24"],"branch_id":null,"executable_argument_node_ids":["LM:S:BL17:location:7-24"],"op":"surface_membership","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM:S:BL17:location:7-24","semantic_operand_id":"LM:S:BL17:location:7-24"}],"quantitative_constraints":[{"relation_type":"surface-landmark","semantic_argument_node_ids":["LM:S:BL17:location:7-24"],"source_statement_id":"S:BL17:location","statement_measurement_ids":["PM:S:BL17:location:00"]},"executable_argument_node_ids":["LM:S:BL17:location:7-24"],"executor":"surface_membership","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM:S:BL17:location:7-24","semantic_operand_id":"LM:S:BL17:location:7-24"}],"semantic_argument_node_ids":["LM:S:BL17:location:7-24"],"semantic_fields":{"bound_measurement_ids":["LM:S:BL17:location:7-24"],"branch_id":null,"op":"surface_membership","quantitative_constraints":[{"relation_type":"surface-landmark","semantic_argument_node_ids":["LM:S:BL17:location:7-24"],"source_statement_id":"S:BL17:location","statement_measurement_ids":["PM:S:BL17:location:00"]},"semantic_fields_hash":"cf359676f8008c132b4047d2f520020bcb7c89a8e7af3f3f72db1f6571444149","status":"RESOLVED"},{"constraint":{"argument_node_ids":["LM:S:BL17:note:1:64-116"],"branch_id":null,"op":"same_level_plane","source_statement_id":"LM:S:BL17:note:1:75-83"},"executable_argument_node_ids":["LM:S:BL17:note:1:64-116"],"executor":"same_level_plane","operand_binding_trace":[{"binding":"child_subfeature","executable_operand_id":"LM:S:BL17:note:1:75-83","semantic_operand_id":"LM:S:BL17:note:1:64-116"}],"reason":"executable operand not uniquely executable","semantic_argument_node_ids":["LM:S:BL17:note:1:64-116"],"semantic_fields":{"bound_measurement_ids":["LM:S:BL17:note:1:64-116"],"branch_id":null,"op":"same_level_plane","quantitative_constraints":[{"relation_type":"same-level","semantic_argument_node_ids":["LM:S:BL17:note:1:64-116"],"source_statement_id":"S:BL17:note:1:75-83","statement_measurement_ids":["PM:S:BL17:note:1:75-83"]},"semantic_fields_hash":"31a6d961162176ebaf9aa3c2d747efedeec7a2953c1f917973bc0d9b90b2f85","status":"UNRESOLVED"}],"status":"UNRESOLVED"}

Coordinates generated: 0 | Legacy C coordinate references: 0

[] WHO source phrase 何者 何者
[] semantic granularity 意味
[] child subfeature 子 実行 意味
[] distinct subfeatures 異なる 実行可能 識別 意味
[] relation operator 関係 演算子 意味
[] condition state 条件 状態
[] RESOLVED/UNRESOLVED 解決済み/未解決
[] ACCEPT
[] REJECT

BL23: RESOLVED -> UNRESOLVED

Defect family: modifier_dependency_loss

WHO source

location | S:BL23:location | page 120 | In the lumbar region, at the same level as the inferior border of the spinous process of the second lumbar vertebra (L2), 1.5 B-cun lateral to the posterior median line.

Semantic decomposition

LM:S:BL23:location:142-168 | source= the posterior median line | class=line.anatomical_line | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"62a4312f07e04de5ebb4791906ed7098bb7c85bea907953c59c8c92bec7d9f4d","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:posterior_median_line","subfeature_type":null} | reason=None

LM:S:BL23:location:43-120 | source=the inferior border of the spinous process of the second lumbar vertebra (L2) | class=bone.bone | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:BL23:location:47-62 | source=inferior border | class=boundary.border | parent=LM:S:BL23:location:66-120 | child=inferior border | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=no approved executable construction rule for canonical subfeature 'inferior_border'

LM:S:BL23:location:66-120 | source=the spinous process of the second lumbar vertebra (L2) | class=bone.bone | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:BL23:location:7-20 | source=lumbar region | class=region.body_region | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"ef7e2ce29e7bdfb2606e98c7c2254ff2a0c99337bcf65192272b78f367ce7244","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:lumbar_region","subfeature_type":null} | reason=None

Composite binding view

PARENT LM:S:BL23:location:66-120 the spinous process of the second lumbar vertebra (L2) status=UNRESOLVED identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:BL23:location:47-62 inferior border status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

Relation execution

RL:S:BL23:location:00:relative-to | op=directional_relation | semantic=['LM:S:BL23:location:142-168'] | executable=['LM:S:BL23:location:142-168'] | dir=lateral | branch=None | source=S:BL23:location | semantic_hash=0b25789777ad668dd609ec872b02fede66284be21bc331bef744a8f8d8a71af7 | status=RESOLVED | reason=None

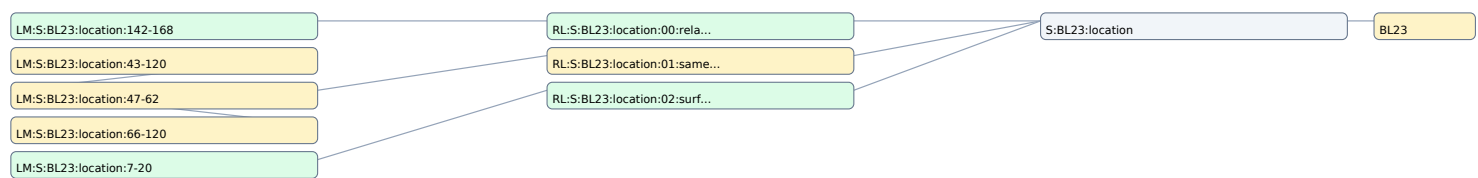
RL:S:BL23:location:01:same-level | op=same_level_plane | semantic=['LM:S:BL23:location:43-120'] | executable=['LM:S:BL23:location:47-62'] | dir=None | branch=None | source=S:BL23:location | semantic_hash=9e1bf3a9cbf0696c0886224691a742fe9f2e7817cfd94f6c3a8e8bb72e5db47 | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:BL23:location:02:surface-landmark | op=surface_membership | semantic=['LM:S:BL23:location:7-20'] | executable=['LM:S:BL23:location:7-20'] | dir=None | branch=None | source=S:BL23:location | semantic_hash=e88774c14962363732cdc9cd35687fa2cd05e8cfe2d32f060cce420b57632f73 | status=RESOLVED | reason=None

Quantitative measurement semantics

{"anchor_landmark_id":"LM:S:BL23:location:142-168","direction":"lateral","executor":"approved_calibration","measurement_id":"PM:S:BL23:location:00","reason":null,"source_statement_id":"S:BL23:location","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":1.5,"who_pdf_page":120}

Dependency graph



Dependency graph edge list

- LM:S:BL23:location:142-168 -> RL:S:BL23:location:00:relative-to [executable_to_relation]
- PM:S:BL23:location:00 -> RL:S:BL23:location:00:relative-to [measurement_to_relation]
- RL:S:BL23:location:00:relative-to -> S:BL23:location [relation_to_statement]
- LM:S:BL23:location:43-120 -> LM:S:BL23:location:47-62 [semantic_to_executable]
- LM:S:BL23:location:47-62 -> RL:S:BL23:location:01:same-level [executable_to_relation]
- RL:S:BL23:location:01:same-level -> S:BL23:location [relation_to_statement]
- LM:S:BL23:location:7-20 -> RL:S:BL23:location:02:surface-landmark [executable_to_relation]

RL:S:BL23:location:02:surface-landmark -> S:BL23:location [relation_to_statement]
LM:S:BL23:location:66-120 -> LM:S:BL23:location:47-62 [parent_to_child_subfeature]
LM:S:BL23:location:142-168 -> PM:S:BL23:location:00 [anchor_to_measurement]
PM:S:BL23:location:00 -> S:BL23:location [measurement_to_statement]
S:BL23:location -> BL23 [statement_to_point]

BEFORE / AFTER diff

BEFORE: {"conditions":[],"relations":[{"constraint":{"argument_node_ids":["LM:S:BL23:location:142-168"],"branch_id":null,"direction":"lateral","op":"directional_relation","source_statement_id":"S:BL23:location"},"executor":"directional_relation","status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:BL23:location:43-120"],"branch_id":null,"op":"same_level_plane","source_statement_id":"S:BL23:location"},"executor":"same_level_plane","status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:BL23:location:7-20"],"branch_id":null,"op":"surface_membership","source_statement_id":"S:BL23:location"},"executor":"surface_membership","status":"RESOLVED"}],"status":"RESOLVED"}

AFTER: {"conditions":[],"relations":[{"constraint":{"argument_node_ids":["LM:S:BL23:location:142-168"],"bound_measurement_ids":["PM:S:BL23:location:00"],"branch_id":null,"direction":"lateral","executable_argument_node_ids":["LM:S:BL23:location:142-168"],"op":"directional_relation","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM:S:BL23:location:142-168","semantic_operand_id":"LM:S:BL23:location:142-168"}],"quantitative_constraints":[{"anchor_landmark_id":"LM:S:BL23:location:142-168","direction":"lateral","kind":"proportional_offset","source_statement_id":"S:BL23:location","source_unit":"B-cun","unit":"B-cun","value":1.5,"who_pdf_page":120}],"relation_type":"relative-to","semantic_argument_node_ids":["LM:S:BL23:location:142-168"],"source_statement_id":"S:BL23:location","statement_measurement_ids":["PM:S:BL23:location:00"]},"executable_argument_node_ids":["LM:S:BL23:location:142-168"],"executor":"directional_relation","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM:S:BL23:location:142-168","semantic_operand_id":"LM:S:BL23:location:142-168"}],"semantic_argument_node_ids":["LM:S:BL23:location:142-168"],"semantic_fields":{"bound_measurement_ids":["PM:S:BL23:location:00"],"branch_id":null,"direction":"lateral","op":"directional_relation","quantitative_constraints":[{"anchor_landmark_id":"LM:S:BL23:location:142-168","direction":"lateral","kind":"proportional_offset","source_statement_id":"S:BL23:location","source_unit":"B-cun","unit":"B-cun","value":1.5,"who_pdf_page":120}],"relation_type":"relative-to","semantic_argument_node_ids":["LM:S:BL23:location:142-168"],"source_statement_id":"S:BL23:location","statement_measurement_ids":["PM:S:BL23:location:00"]},"semantic_fields_hash":"0b25789777ad668dd609ec872b02fed66284be21bc331bef744a8f8d8a71af7","status":"RESOLVED"}, {"executable_argument_node_ids":["LM:S:BL23:location:47-62"],"executor":"same_level_plane","operand_binding_trace":[{"binding":"child_subfeature","executable_operand_id":"LM:S:BL23:location:47-62","semantic_operand_id":"LM:S:BL23:location:43-120"}],"reason":"executable operand not uniquely executable","semantic_argument_node_ids":["LM:S:BL23:location:43-120"],"semantic_fields":{"bound_measurement_ids":["PM:S:BL23:location:00"],"branch_id":null,"op":"same_level_plane","quantitative_constraints":[{"relation_type":"same-level","semantic_argument_node_ids":["LM:S:BL23:location:43-120"],"source_statement_id":"S:BL23:location","statement_measurement_ids":["PM:S:BL23:location:00"]},"semantic_fields_hash":"9e1bf3a9cbf0696c0886224691a742fe9f2e7817cddf94f6c3a8e8bb72e5db47","status":"UNRESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:BL23:location:7-20"],"bound_measurement_ids":["PM:S:BL23:location:00"],"branch_id":null,"executable_argument_node_ids":["LM:S:BL23:location:7-20"],"op":"surface_membership","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM:S:BL23:location:7-20","semantic_operand_id":"LM:S:BL23:location:7-20"}],"quantitative_constraints":[{"relation_type":"surface-landmark","semantic_argument_node_ids":["LM:S:BL23:location:7-20"],"source_statement_id":"S:BL23:location","statement_measurement_ids":["PM:S:BL23:location:00"]},"semantic_fields_hash":"e88774c14962363732cdc9cd35687fa2cd05e8cfe2d32f060cce420b57632f73","status":"RESOLVED"}],"status":"UNRESOLVED"}

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature
- [] distinct subfeatures
- [] relation operator
- [] condition state
- [] RESOLVED/UNRESOLVED
- [] ACCEPT
- [] REJECT

BL25: RESOLVED -> UNRESOLVED

Defect family: modifier_dependency_loss

WHO source

location | S:BL25:location | page 121 | In the lumbar region, at the same level as the inferior border of the spinous process of the fourth lumbar vertebra (L4), 1.5 B-cun lateral to the posterior median line.

Semantic decomposition

LM:S:BL25:location:142-168 | source= the posterior median line | class=line.anatomical_line | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"62a4312f07e04de5ebb4791906ed7098bb7c85bea907953c59c8c92bec7d9f4d","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:posterior_median_line","subfeature_type":null} | reason=None

LM:S:BL25:location:43-120 | source=the inferior border of the spinous process of the fourth lumbar vertebra (L4) | class=bone.bone | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:BL25:location:47-62 | source=inferior border | class=boundary.border | parent=LM:S:BL25:location:66-120 | child=inferior border | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=no approved executable construction rule for canonical subfeature 'inferior_border'

LM:S:BL25:location:66-120 | source=the spinous process of the fourth lumbar vertebra (L4) | class=bone.bone | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:BL25:location:7-20 | source=lumbar region | class=region.body_region | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"ef7e2ce29e7bdfb2606e98c7c2254ff2a0c99337bcf65192272b78f367ce7244","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:lumbar_region","subfeature_type":null} | reason=None

Composite binding view

PARENT LM:S:BL25:location:66-120 the spinous process of the fourth lumbar vertebra (L4) status=UNRESOLVED
identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:BL25:location:47-62 inferior border status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

Relation execution

RL:S:BL25:location:00:relative-to | op=directional_relation | semantic=['LM:S:BL25:location:142-168'] | executable=['LM:S:BL25:location:142-168'] | dir=lateral | branch=None | source=S:BL25:location | semantic_hash=a8202d3c8d87b2088b30dcf634c5b4e1c45bf6bfa4a7678e5770c2ba90d91649 | status=RESOLVED | reason=None

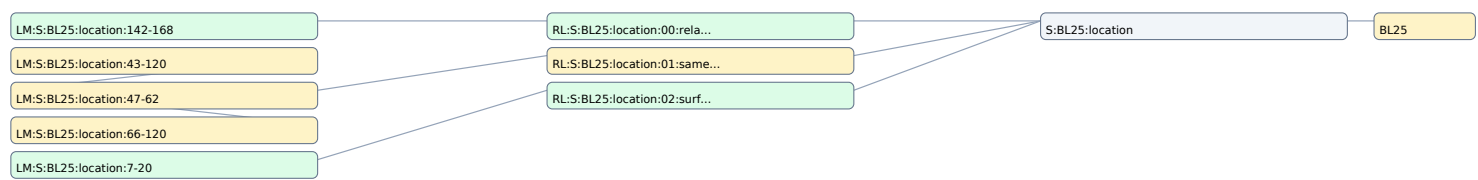
RL:S:BL25:location:01:same-level | op=same_level_plane | semantic=['LM:S:BL25:location:43-120'] | executable=['LM:S:BL25:location:47-62'] | dir=None | branch=None | source=S:BL25:location | semantic_hash=805eaacd77a202b229db0b93db4fd57760dfdfb71faacb8d5f90153e328de187 | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:BL25:location:02:surface-landmark | op=surface_membership | semantic=['LM:S:BL25:location:7-20'] | executable=['LM:S:BL25:location:7-20'] | dir=None | branch=None | source=S:BL25:location | semantic_hash=aa7807c05074f5ec9bef13d28a65d0f056a4081135f8a63404b15bf17f009aeb | status=RESOLVED | reason=None

Quantitative measurement semantics

{"anchor_landmark_id":"LM:S:BL25:location:142-168","direction":"lateral","executor":"approved_calibration","measurement_id":"PM:S:BL25:location:00","reason":null,"source_statement_id":"S:BL25:location","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":1.5,"who_pdf_page":121}

Dependency graph



Dependency graph edge list

- LM:S:BL25:location:142-168 -> RL:S:BL25:location:00:relative-to [executable_to_relation]
- PM:S:BL25:location:00 -> RL:S:BL25:location:00:relative-to [measurement_to_relation]
- RL:S:BL25:location:00:relative-to -> S:BL25:location [relation_to_statement]
- LM:S:BL25:location:43-120 -> LM:S:BL25:location:47-62 [semantic_to_executable]
- LM:S:BL25:location:47-62 -> RL:S:BL25:location:01:same-level [executable_to_relation]
- RL:S:BL25:location:01:same-level -> S:BL25:location [relation_to_statement]
- LM:S:BL25:location:7-20 -> RL:S:BL25:location:02:surface-landmark [executable_to_relation]

RL:S:BL25:location:02:surface-landmark -> S:BL25:location [relation_to_statement]
LM:S:BL25:location:66-120 -> LM:S:BL25:location:47-62 [parent_to_child_subfeature]
LM:S:BL25:location:142-168 -> PM:S:BL25:location:00 [anchor_to_measurement]
PM:S:BL25:location:00 -> S:BL25:location [measurement_to_statement]
S:BL25:location -> BL25 [statement_to_point]

BEFORE / AFTER diff

BEFORE: {"conditions":[],"relations":[{"constraint":{"argument_node_ids":["LM:S:BL25:location:142-168"],"branch_id":null,"direction":"lateral","op":"directional_relation","source_statement_id":"S:BL25:location"},"executor":"directional_relation","status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:BL25:location:43-120"],"branch_id":null,"op":"same_level_plane","source_statement_id":"S:BL25:location"},"executor":"same_level_plane","status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:BL25:location:7-20"],"branch_id":null,"op":"surface_membership","source_statement_id":"S:BL25:location"},"executor":"surface_membership","status":"RESOLVED"}],"status":"RESOLVED"}

AFTER: {"conditions":[],"relations":[{"constraint":{"argument_node_ids":["LM:S:BL25:location:142-168"],"bound_measurement_ids":["PM:S:BL25:location:00"],"branch_id":null,"direction":"lateral","executable_argument_node_ids":["LM:S:BL25:location:142-168"],"op":"directional_relation","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM:S:BL25:location:142-168","semantic_operand_id":"LM:S:BL25:location:142-168"}],"quantitative_constraints":[{"anchor_landmark_id":"LM:S:BL25:location:142-168","direction":"lateral","kind":"proportional_offset","source_statement_id":"S:BL25:location","source_unit":"B-cun","unit":"B-cun","value":1.5,"who_pdf_page":121}],"relation_type":"relative-to","semantic_argument_node_ids":["LM:S:BL25:location:142-168"],"source_statement_id":"S:BL25:location","statement_measurement_ids":["PM:S:BL25:location:00"]},"executable_argument_node_ids":["LM:S:BL25:location:142-168"],"executor":"directional_relation","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM:S:BL25:location:142-168","semantic_operand_id":"LM:S:BL25:location:142-168"}],"semantic_argument_node_ids":["LM:S:BL25:location:142-168"],"semantic_fields":{"bound_measurement_ids":["PM:S:BL25:location:00"],"branch_id":null,"direction":"lateral","op":"directional_relation","quantitative_constraints":[{"anchor_landmark_id":"LM:S:BL25:location:142-168","direction":"lateral","kind":"proportional_offset","source_statement_id":"S:BL25:location","source_unit":"B-cun","unit":"B-cun","value":1.5,"who_pdf_page":121}],"relation_type":"relative-to","semantic_argument_node_ids":["LM:S:BL25:location:142-168"],"source_statement_id":"S:BL25:location","statement_measurement_ids":["PM:S:BL25:location:00"]},"semantic_fields_hash":"a8202d3c8d87b2088b30dcf634c5b4e1c45bf6bfa4a7678e5770c2ba90d91649","status":"RESOLVED"}, {"executable_argument_node_ids":["LM:S:BL25:location:47-62"],"op":"same_level_plane","operand_binding_trace":[{"binding":"child_subfeature","executable_operand_id":"LM:S:BL25:location:47-62","semantic_operand_id":"LM:S:BL25:location:43-120"}],"reason":"executable operand not uniquely executable","semantic_argument_node_ids":["LM:S:BL25:location:43-120"],"semantic_fields":{"bound_measurement_ids":[],"branch_id":null,"op":"same_level_plane","quantitative_constraints":[],"relation_type":"same-level","semantic_argument_node_ids":["LM:S:BL25:location:43-120"],"source_statement_id":"S:BL25:location","statement_measurement_ids":["PM:S:BL25:location:00"]},"semantic_fields_hash":"805eaacd77a202b229db0b93db4fd57760fdfbb71faacb8d5f90153e328de187","status":"UNRESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:BL25:location:7-20"],"bound_measurement_ids":[],"branch_id":null,"executable_argument_node_ids":["LM:S:BL25:location:7-20"],"op":"surface_membership","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM:S:BL25:location:7-20","semantic_operand_id":"LM:S:BL25:location:7-20"}],"quantitative_constraints":[],"relation_type":"surface-landmark","semantic_argument_node_ids":["LM:S:BL25:location:7-20"],"source_statement_id":"S:BL25:location","statement_measurement_ids":["PM:S:BL25:location:00"]},"executable_argument_node_ids":["LM:S:BL25:location:7-20"],"executor":"surface_membership","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM:S:BL25:location:7-20","semantic_operand_id":"LM:S:BL25:location:7-20"}],"semantic_argument_node_ids":["LM:S:BL25:location:7-20"],"semantic_fields":{"bound_measurement_ids":[],"branch_id":null,"op":"surface_membership","quantitative_constraints":[],"relation_type":"surface-landmark","semantic_argument_node_ids":["LM:S:BL25:location:7-20"],"source_statement_id":"S:BL25:location","statement_measurement_ids":["PM:S:BL25:location:00"]},"semantic_fields_hash":"a7807c05074f5ec9bef13d28a65d0f056a4081135f8a63404b15bf17f009aeb","status":"RESOLVED"},"status":"UNRESOLVED"}

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase 谁 什么 哪里
- [] semantic granularity 语义
- [] child subfeature 子特征 execution 执行
- [] distinct subfeatures 独特可执行身份 区别
- [] relation operator 关系 操作 操作
- [] condition state 条件 状态
- [] RESOLVED/UNRESOLVED 解决 未解决
- [] ACCEPT
- [] REJECT

ST9: RESOLVED -> UNRESOLVED

Defect family: modifier_dependency_loss

WHO source

location | S:ST9:location | page 59 | In the anterior region of the neck, at the same level as the superior border of the thyroid cartilage, anterior to the sterno-cleidomastoid muscle, over the common carotid artery.

note | S:ST9:note:1 | page 59 | Note 1: The sternocleidomastoid muscle is more distinct when the head is turned to the opposite side against resistance.

note | S:ST9:note:2 | page 59 | Note 2: ST9 is located at the same level as LI18, SI16 and the superior border of the thyroid cartilage. St9 is located anterior to the sternocleidomastoid muscle, and SI16, posterior to the sternocleidomastoid muscle and LI18, between the anterior and pos-terior borders of the sternocleidomastoid muscle.

Semantic decomposition

LM:S:ST9:location:114-146 | source= the sterno-cleidomastoid muscle | class=muscle | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA13407","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:ST9:location:157-178 | source=common carotid artery | class=vessel | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA3939","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:ST9:location:57-101 | source=the superior border of the thyroid cartilage | class=cartilage | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:ST9:location:61-76 | source=superior border | class=boundary.border | parent=LM:S:ST9:location:84-101 | child=superior border | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=no approved executable construction rule for canonical subfeature 'superior_border'

LM:S:ST9:location:7-34 | source=anterior region of the neck | class=region.body_region | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"71f7a5ab627b3c93a9aceb574aabdb27ca3577688316b7de105a5effa7840dc7","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:anterior_neck","subfeature_type":null} | reason=None

LM:S:ST9:location:84-101 | source=thyroid cartilage | class=cartilage | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA55099","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:ST9:note:2:131-162 | source= the sternocleidomastoid muscle | class=muscle | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA13407","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:ST9:note:2:186-226 | source= the sternocleidomastoid muscle and LI18 | class=muscle | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA13407","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:ST9:note:2:240-305:6f7a364e | source=anterior and pos-terior borders of the sternocleidomastoid muscle | class=muscle | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:ST9:note:2:253-305:a442a1a4 | source=pos-terior borders of the sternocleidomastoid muscle | class=muscle | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:ST9:note:2:44-48 | source=LI18 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"LI18","registry_id":null,"subfeature_type":null} | reason=None

LM:S:ST9:note:2:63-78 | source=superior border | class=boundary.border | parent=LM:S:ST9:note:2:82-103 | child=superior border | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=no approved executable construction rule for canonical subfeature 'superior_border'

LM:S:ST9:note:2:82-103 | source=the thyroid cartilage | class=cartilage | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA55099","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM21:S:ST9:note:2:50-54:6775aaafef | source=SI16 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"SI16","registry_id":null,"subfeature_type":null} | reason=None

LM21:S:ST9:note:2:168-172:6775aaafef | source=SI16 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"SI16","registry_id":null,"subfeature_type":null} | reason=None

LM21:S:ST9:note:2:222-226:34574dfae8 | source=LI18 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"LI18","registry_id":null,"subfeature_type":null} | reason=None

Composite binding view

PARENT LM:S:ST9:location:84-101 thyroid cartilage status=RESOLVED identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA55099","geometry_hash":null,"kind":"","fma_concept":"","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:ST9:location:61-76 superior border status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

PARENT LM:S:ST9:note:2:82-103 the thyroid cartilage status=RESOLVED identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA55099","geometry_hash":null,"kind":"","fma_concept":"","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:ST9:note:2:63-78 superior border status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

Relation execution

RL:S:ST9:location:00:relative-to | op=directional_relation | semantic=['LM:S:ST9:location:114-146'] | executable=['LM:S:ST9:location:114-146'] | dir=anterior | branch=None | source=S:ST9:location | semantic_hash=b9f8325fb6ae3238a99897595686ff6ef885a38c952379c6e5bf21d896cc841e | status=RESOLVED | reason=None

RL:S:ST9:location:01:same-level | op=same_level_plane | semantic=['LM:S:ST9:location:57-101'] | executable=['LM:S:ST9:location:61-76'] | dir=None | branch=None | source=S:ST9:location | semantic_hash=f1d571cb655e425a5b28acd8a07d1a99afe1b906dc57e5f0df090f6aac17957a | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:ST9:location:02:overlies | op=overlies_projection | semantic=['LM:S:ST9:location:157-178'] | executable=['LM:S:ST9:location:157-178'] | dir=None | branch=None | source=S:ST9:location | semantic_hash=65cdfd954c54cfcc84dfbf36b8bfa4568fec90fba1fba67782e544034ad803e7 | status=RESOLVED | reason=None

RL:S:ST9:location:03:surface-landmark | op=surface_membership | semantic=['LM:S:ST9:location:7-34'] | executable=['LM:S:ST9:location:7-34'] | dir=None | branch=None | source=S:ST9:location | semantic_hash=13aa4c3afb628dd0ac129ed1faf2931fd5999687f0cc7d4defbadddd3cc5dbcd | status=RESOLVED | reason=None

RL:S:ST9:note:2:00:between | op=between_constraint | semantic=['LM:S:ST9:note:2:240-305:6f7a364e', 'LM:S:ST9:note:2:253-305:a442a1a4'] | executable=[] | dir=None | branch=None | source=S:ST9:note:2 | semantic_hash=31b2630063f9b57071ffd6899f519f9ab8963309820b113b9e927a15fb88764f | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:ST9:note:2:01:relative-to | op=directional_relation | semantic=['LM:S:ST9:note:2:131-162'] | executable=['LM:S:ST9:note:2:131-162'] | dir=anterior | branch=None | source=S:ST9:note:2 | semantic_hash=6c8185584e90fd0e4cdb316e366ad7189ab79f80a98d898e258cb3c755c3752e | status=RESOLVED | reason=None

RL:S:ST9:note:2:02:relative-to | op=directional_relation | semantic=['LM:S:ST9:note:2:186-226'] | executable=['LM:S:ST9:note:2:186-226'] | dir=posterior | branch=None | source=S:ST9:note:2 | semantic_hash=5743720b41ad303c0e0c098b41c34bf537246a45ac2a7d47eeccb6ee3f12c3e6 | status=RESOLVED | reason=None

RL:S:ST9:note:2:03:same-level | op=same_level_plane | semantic=['LM:S:ST9:note:2:44-48'] | executable=['LM:S:ST9:note:2:44-48'] | dir=None | branch=None | source=S:ST9:note:2 | semantic_hash=13d6e0b895c93f89582165316f8ca947dee518d35d1777227b62480c5479efcb | status=RESOLVED | reason=None

RL21:S:ST9:note:2:44-48:reference-acupoint:dfa4bc85d2 | op=reference_dependency | semantic=['LM:S:ST9:note:2:44-48'] | executable=['LM:S:ST9:note:2:44-48'] | dir=None | branch=None | source=S:ST9:note:2 | semantic_hash=81ab938ff14783315d59a9607cd2359f2aca4d4e474a3dd45dcfbfb1d79c6883 | status=RESOLVED | reason=None

RL21:S:ST9:note:2:50-54:reference-acupoint:cf7c9e057c | op=reference_dependency | semantic=['LM21:S:ST9:note:2:50-54:6775aaafeb'] | executable=['LM21:S:ST9:note:2:50-54:6775aaafeb'] | dir=None | branch=None | source=S:ST9:note:2 | semantic_hash=54039e732eaae54438ba16eac9423bad4e40e74e941338615d245cb368c47717 | status=RESOLVED | reason=None

RL21:S:ST9:note:2:168-172:reference-acupoint:a3b2f0d073 | op=reference_dependency | semantic=['LM21:S:ST9:note:2:168-172:6775aaafeb'] | executable=['LM21:S:ST9:note:2:168-172:6775aaafeb'] | dir=None | branch=None | source=S:ST9:note:2 | semantic_hash=76177c0c0749865756dd84b12fc968be593a2c07d3b7e2c3a80d6824237d9aa9 | status=RESOLVED | reason=None

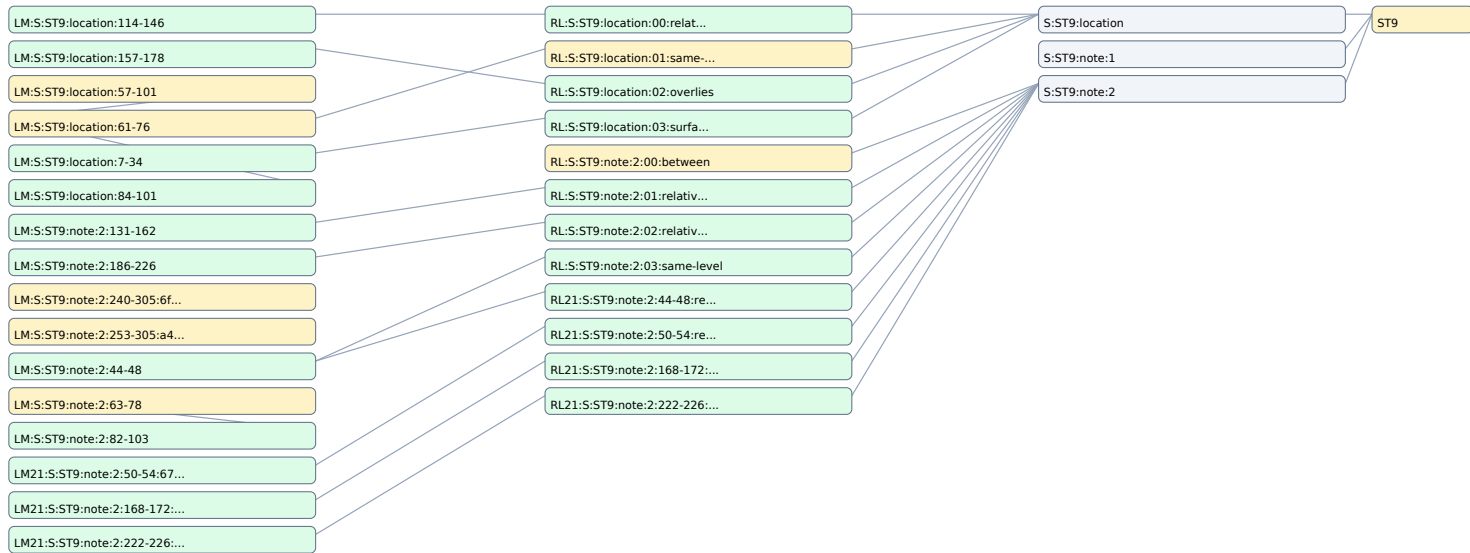
RL21:S:ST9:note:2:222-226:reference-acupoint:01273086b9 | op=reference_dependency | semantic=['LM21:S:ST9:note:2:222-226:34574dfae8'] | executable=['LM21:S:ST9:note:2:222-226:34574dfae8'] | dir=None | branch=None | source=S:ST9:note:2 | semantic_hash=f5ccf40bde1ae23e6790d8bfd879bce945126b8bd8c27f47fd557915905e595 | status=RESOLVED | reason=None

Conditions

{"branch_id":"branch_1","condition_id":"CD:S:ST9:note:1:00","condition_type":"body_position","default_pose_compatible":false,"reason":"condition requires pose incompatible with default BodyParts3D pose","repaired_status":"CONDITIONAL","source_raw":"when the head is turned to the opposite side against resistance","source_statement_id":"S:ST9:note:1"}

{"branch_id":"branch_2","condition_id":"CD:S:ST9:note:1:01","condition_type":"palpation_dependent","default_pose_compatible":null,"reason":"condition preserved as non-executed conditional branch","repaired_status":"CONDITIONAL","source_raw":"more distinct when the head is turned to the opposite side against resistance","source_statement_id":"S:ST9:note:1"}

Dependency graph



Dependency graph edge list

LM:S:ST9:location:114-146 -> RL:S:ST9:location:00:relative-to [executable_to_relation]

RL:S:ST9:location:00:relative-to -> S:ST9:location [relation_to_statement]

LM:S:ST9:location:57-101 -> LM:S:ST9:location:61-76 [semantic_to_executable]

LM:S:ST9:location:61-76 -> RL:S:ST9:location:01:same-level [executable_to_relation]

RL:S:ST9:location:01:same-level -> S:ST9:location [relation_to_statement]

LM:S:ST9:location:157-178 -> RL:S:ST9:location:02:overlies [executable_to_relation]

RL:S:ST9:location:02:overlies -> S:ST9:location [relation_to_statement]

LM:S:ST9:location:7-34 -> RL:S:ST9:location:03:surface-landmark [executable_to_relation]

RL:S:ST9:location:03:surface-landmark -> S:ST9:location [relation_to_statement]

RL:S:ST9:note:2:00:between -> S:ST9:note:2 [relation_to_statement]

LM:S:ST9:note:2:131-162 -> RL:S:ST9:note:2:01:relative-to [executable_to_relation]

RL:S:ST9:note:2:01:relative-to -> S:ST9:note:2 [relation_to_statement]

LM:S:ST9:note:2:186-226 -> RL:S:ST9:note:2:02:relative-to [executable_to_relation]

RL:S:ST9:note:2:02:relative-to -> S:ST9:note:2 [relation_to_statement]

LM:S:ST9:note:2:44-48 -> RL:S:ST9:note:2:03:same-level [executable_to_relation]

RL:S:ST9:note:2:03:same-level -> S:ST9:note:2 [relation_to_statement]

LM:S:ST9:note:2:44-48 -> RL21:S:ST9:note:2:44-48:reference-acupoint:dfa4bc85d2 [executable_to_relation]

RL21:S:ST9:note:2:44-48:reference-acupoint:dfa4bc85d2 -> S:ST9:note:2 [relation_to_statement]

LM21:S:ST9:note:2:50-54:6775aaafeb -> RL21:S:ST9:note:2:50-54:reference-acupoint:cf7c9e057c [executable_to_relation]

RL21:S:ST9:note:2:50-54:reference-acupoint:cf7c9e057c -> S:ST9:note:2 [relation_to_statement]

LM21:S:ST9:note:2:168-172:6775aaafeb -> RL21:S:ST9:note:2:168-172:reference-acupoint:a3b2f0d073 [executable_to_relation]

RL21:S:ST9:note:2:168-172:reference-acupoint:a3b2f0d073 -> S:ST9:note:2 [relation_to_statement]

LM21:S:ST9:note:2:222-226:34574dfae8 -> RL21:S:ST9:note:2:222-226:reference-acupoint:01273086b9 [executable_to_relation]

RL21:S:ST9:note:2:222-226:reference-acupoint:01273086b9 -> S:ST9:note:2 [relation_to_statement]

LM:S:ST9:location:84-101 -> LM:S:ST9:location:61-76 [parent_to_child_subfeature]

LM:S:ST9:note:2:82-103 -> LM:S:ST9:note:2:63-78 [parent_to_child_subfeature]

S:ST9:location -> ST9 [statement_to_point]

S:ST9:note:1 -> ST9 [statement_to_point]

S:ST9:note:2 -> ST9 [statement_to_point]

BEFORE / AFTER diff

BEFORE: {"conditions":[{"branch_id":"branch_1","condition_type":"body_position","executor":"condition_contract","reason":"condition requires pose incompatible with default BodyParts3D pose","status":"CONDITIONAL"}, {"branch_id":"branch_2","condition_type":"palpation_dependent","executor":"condition_contract","reason":"branch preserved; not forced into default-pose primary computation","status":"CONDITIONAL"}], "relations":[{"constraint":{"argument_node_ids":["LM:S:ST9:location:114-146"],"branch_id":null,"direction":"anterior","op":"directional_relation","source_statement_id":"S:ST9:location"},"executor":"directional_relation","status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:ST9:location:57-101"],"branch_id":null,"op":"same_level_plane","source_statement_id":"S:ST9:location"},"executor":"same_level_plane","status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:ST9:location:157-178"],"branch_id":null,"op":"overlies_projection","source_statement_id":"S:ST9:location"},"executor":"overlies_projection","status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:ST9:location:7-34"],"branch_id":null,"op":"surface_membership","source_statement_id":"S:ST9:location"},"executor":"surface_membership","status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:S:ST9:note:2:240-305:6f7a364e","LM:S:ST9:note:2:253-305:a442a1a4"],"branch_id":null,"op":"between_constraint","source_statement_id":"S:ST9:note:2"},"executor":"between_constraint","status":"RESOLVED"}, {"constraint":{"argument_node_ids":["LM:

statement_id":"S:ST9:note:2","statement_measurement_ids":[],"semantic_fields_hash":"54039e732eaae54438ba16eac9423bad4e40e74e941338615d245cb368c47717","status":"RESOLVED"},{"constraint":{"argument_node_ids":["LM21:S:ST9:note:2:168-172:6775aaafeb"],"bound_measurement_ids":[],"branch_id":null,"executable_argument_node_ids":["LM21:S:ST9:note:2:168-172:6775aaafeb"],"op":"reference_dependency","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM21:S:ST9:note:2:168-172:6775aaafeb","semantic_operand_id":"LM21:S:ST9:note:2:168-172:6775aaafeb"}],"quantitative_constraints":[],"relation_type":"reference-acupoint","semantic_argument_node_ids":["LM21:S:ST9:note:2:168-172:6775aaafeb"],"source_statement_id":"S:ST9:note:2","statement_measurement_ids":[],"executable_argument_node_ids":["LM21:S:ST9:note:2:168-172:6775aaafeb"],"executor":"reference_dependency","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM21:S:ST9:note:2:168-172:6775aaafeb","semantic_operand_id":"LM21:S:ST9:note:2:168-172:6775aaafeb"}],"semantic_argument_node_ids":["LM21:S:ST9:note:2:168-172:6775aaafeb"],"semantic_fields":{"bound_measurement_ids":[],"branch_id":null,"op":"reference_dependency","quantitative_constraints":[],"relation_type":"reference-acupoint","semantic_argument_node_ids":["LM21:S:ST9:note:2:168-172:6775aaafeb"],"source_statement_id":"S:ST9:note:2","statement_measurement_ids":[],"semantic_fields_hash":"76177c0c0749865756dd84b12fc968be593a2c07d3b7e2c3a80d6824237d9aa9","status":"RESOLVED"},{"constraint":{"argument_node_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"bound_measurement_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"branch_id":null,"executable_argument_node_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"op":"reference_dependency","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM21:S:ST9:note:2:222-226:34574dfae8","semantic_operand_id":"LM21:S:ST9:note:2:222-226:34574dfae8"}],"quantitative_constraints":[],"relation_type":"reference-acupoint","semantic_argument_node_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"source_statement_id":"S:ST9:note:2","statement_measurement_ids":[],"executable_argument_node_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"executor":"reference_dependency","operand_binding_trace":[{"binding":"identity","executable_operand_id":"LM21:S:ST9:note:2:222-226:34574dfae8","semantic_operand_id":"LM21:S:ST9:note:2:222-226:34574dfae8"}],"semantic_argument_node_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"semantic_fields":{"bound_measurement_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"branch_id":null,"op":"reference_dependency","quantitative_constraints":[],"relation_type":"reference-acupoint","semantic_argument_node_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"source_statement_id":"S:ST9:note:2","statement_measurement_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"semantic_fields_hash":"f5ccf40bde1ae23e6790d8bfd879bce945126b8bd8c27f47fd557915905e595","status":"RESOLVED"}],"status":"UNRESOLVED"}

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature
- [] distinct subfeatures
- [] relation operator
- [] condition state
- [] RESOLVED/UNRESOLVED
- [] ACCEPT
- [] REJECT

TE20: UNRESOLVED -> UNRESOLVED

Defect family: condition_semantics_loss

WHO source

location | S:TE20:location | page 176 | On the head, just superior to the auricular apex.

note | S:TE20:note:1 | page 176 | Note: With the auricle folded forward and pressed against the head, the point is located where the auricular apex touches the head.

Semantic decomposition

LM:S:TE20:location:29-48 | source= the auricular apex | class=soft_tissue_feature | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={ "concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:TE20:location:7-11 | source=head | class=region.body_region | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={ "concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA7154","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

Composite binding view

Relation execution

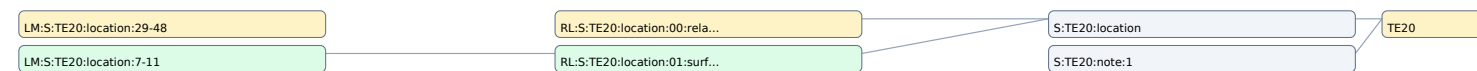
RL:S:TE20:location:00:relative-to | op=directional_relation | semantic=['LM:S:TE20:location:29-48'] | executable=[] | dir=superior | branch=None | source=S:TE20:location | semantic_hash=5e7c940e23aa782af7b50e1412d337e17cf2989f97b00c462063c8ac30ed672e | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:TE20:location:01:surface-landmark | op=surface_membership | semantic=['LM:S:TE20:location:7-11'] | executable=['LM:S:TE20:location:7-11'] | dir=None | branch=None | source=S:TE20:location | semantic_hash=81a23a3a40d0239b756475a302395484815191a0953490656f902075af3530df | status=RESOLVED | reason=None

Conditions

{ "branch_id":"branch_1","condition_id":"CD:S:TE20:note:1:00","condition_type":"body_position","default_pose_compatible":false,"reason":"condition requires pose incompatible with default BodyParts3D pose","repaired_status":"CONDITIONAL","source_raw":"With the auricle folded","source_statement_id":"S:TE20:note:1" }

Dependency graph



Dependency graph edge list

RL:S:TE20:location:00:relative-to -> S:TE20:location [relation_to_statement]

LM:S:TE20:location:7-11 -> RL:S:TE20:location:01:surface-landmark [executable_to_relation]

RL:S:TE20:location:01:surface-landmark -> S:TE20:location [relation_to_statement]

S:TE20:location -> TE20 [statement_to_point]

S:TE20:note:1 -> TE20 [statement_to_point]

BEFORE / AFTER diff

BEFORE: { "conditions": [{ "branch_id": "branch_1", "condition_type": "body_position", "executor": "condition_contract", "reason": null, "status": "RESOLVED" }], "relations": [{ "argument_node_ids": ["LM:S:TE20:location:29-48"], "executor": "directional_relation", "reason": "argument landmark not uniquely executable", "status": "UNRESOLVED" }, { "constraint": { "argument_node_ids": ["LM:S:TE20:location:7-11"], "branch_id": null, "op": "surface_membership", "source_statement_id": "S:TE20:location" }, "executor": "surface_membership", "status": "RESOLVED" }, { "status": "UNRESOLVED" }] }

AFTER: { "conditions": [{ "branch_id": "branch_1", "condition_type": "body_position", "default_pose_compatible": false, "executor": "condition_contract", "reason": "condition requires pose incompatible with default BodyParts3D pose", "source_span": { "char_end": 29, "char_start": 6, "condition_type": "body_position", "role": null, "source_raw": "With the auricle folded" }, "source_statement_id": "S:TE20:note:1", "status": "CONDITIONAL" }], "relations": [{ "executable_argument_node_ids": ["LM:S:TE20:location:29-48"], "reason": "semantic composite has multiple child subfeatures and no unique executable operand", "semantic_argument_node_ids": ["LM:S:TE20:location:29-48"], "semantic_fields": { "bound_measurement_ids": [], "branch_id": null, "direction": "superior", "op": "directional_relation", "quantitative_constraints": [], "relation_type": "relative-to", "semantic_argument_node_ids": ["LM:S:TE20:location:29-48"], "source_statement_id": "S:TE20:location", "statement_measurement_ids": [], "semantic_fields_hash": "5e7c940e23aa782af7b50e1412d337e17cf2989f97b00c462063c8ac30ed672e", "status": "UNRESOLVED" }, { "constraint": { "argument_node_ids": ["LM:S:TE20:location:7-11"], "bound_measurement_ids": [], "branch_id": null, "executable_argument_node_ids": ["LM:S:TE20:location:7-11"], "op": "surface_membership", "operand_binding_trace": [{ "binding": "identity", "executable_operand_id": "LM:S:TE20:location:7-11", "semantic_operand_id": "LM:S:TE20:location:7-11", "semantic_operand_id": "LM:S:TE20:location:7-11" }, { "binding": "missing_executable_subfeature", "executable_operand_id": [], "semantic_operand_id": "LM:S:TE20:location:7-11", "semantic_operand_id": "LM:S:TE20:location:7-11" }], "quantitative_constraints": [], "relation_type": "surface-landmark", "semantic_argument_node_ids": ["LM:S:TE20:location:7-11"], "source_statement_id": "S:TE20:location", "statement_measurement_ids": [], "executable_argument_node_ids": ["LM:S:TE20:location:7-11"], "executor": "surface_membership", "operand_binding_trace": [{ "binding": "identity", "executable_operand_id": "LM:S:TE20:location:7-11", "semantic_operand_id": "LM:S:TE20:location:7-11", "semantic_operand_id": "LM:S:TE20:location:7-11" }, { "binding": "missing_executable_subfeature", "executable_operand_id": [], "semantic_operand_id": "LM:S:TE20:location:7-11", "semantic_operand_id": "LM:S:TE20:location:7-11" }], "quantitative_constraints": [], "relation_type": "surface-landmark", "semantic_argument_node_ids": ["LM:S:TE20:location:7-11"], "source_statement_id": "S:TE20:location", "statement_measurement_ids": [], "semantic_fields_hash": "81a23a3a40d0239b756475a302395484815191a0953490656f902075af3530df", "status": "RESOLVED" }, { "status": "UNRESOLVED" }] }

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature execution
- [] distinct subfeatures distinct executable identity
- [] relation operator operand

[] condition state
[] RESOLVED/UNRESOLVED
[] ACCEPT
[] REJECT

HT7: UNRESOLVED -> UNRESOLVED

Defect family: relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, direction_preservation_failure, relation_semantic_hash_mismatch, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete

WHO source

location | S:HT7:location | page 94 | On the anteromedial aspect of the wrist, radial to the flexor carpi ulnaris tendon, on the palmar wrist crease.

note | S:HT7:note:1 | page 94 | Note: In the depression radial to the proximal border of the pisiform bone, on the palmar wrist crease.

Semantic decomposition

LM:S:HT7:location:50-82 | source= the flexor carpi ulnaris tendon | class=tendon | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:HT7:location:7-26 | source=anteromedial aspect | class=surface.aspect | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"6621b519a3e9551fb3372da3e363c6dd82762c2bc01e5e66f461ac31124c6054","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:anteromedial_wrist","subfeature_type":null} | reason=None

LM:S:HT7:location:91-110 | source=palmar wrist crease | class=crease.skin_crease | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"383e200f286d70d3c13689830673e15bb470bf299882294478ffe17ce465fa2f","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:palmar_wrist_crease","subfeature_type":null} | reason=None

LM:S:HT7:note:1:13-23 | source=depression | class=fossa_or_depression | parent=None | child=None | executor=explicit_unresolved_taxonomy | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=ambiguous_generic

LM:S:HT7:note:1:33-75 | source= the proximal border of the pisiform bone | class=boundary.border | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:HT7:note:1:38-54 | source=proximal border | class=boundary.border | parent=LM:S:HT7:note:1:58-75 | child=proximal border | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=parent entity is not uniquely resolved

LM:S:HT7:note:1:58-75 | source=the pisiform bone | class=bone.bone | parent=None | child=None | executor=atlas_name_resolution | status=MULTIPLE | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=multiple atlas-name candidates

LM:S:HT7:note:1:84-103 | source=palmar wrist crease | class=crease.skin_crease | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"383e200f286d70d3c13689830673e15bb470bf299882294478ffe17ce465fa2f","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:palmar_wrist_crease","subfeature_type":null} | reason=None

Composite binding view

PARENT LM:S:HT7:note:1:58-75 the pisiform bone status=MULTIPLE identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:HT7:note:1:38-54 proximal border status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

Relation execution

RL:S:HT7:location:00:relative-to | op=directional_relation | semantic=['LM:S:HT7:location:50-82'] | executable=[] | dir=radial | branch=None | source=S:HT7:location | semantic_hash=3d7a2492d0838f8d685b5e84c29b01499b7792f0a24e8291281969fca1b38bd4 | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:HT7:location:01:surface-landmark | op=surface_membership | semantic=['LM:S:HT7:location:7-26'] | executable=['LM:S:HT7:location:7-26'] | dir=None | branch=None | source=S:HT7:location | semantic_hash=71873a6f9589d55db0fbc83821b943b957f3fc33ee0b6482e2ca56300c9fe410 | status=RESOLVED | reason=None

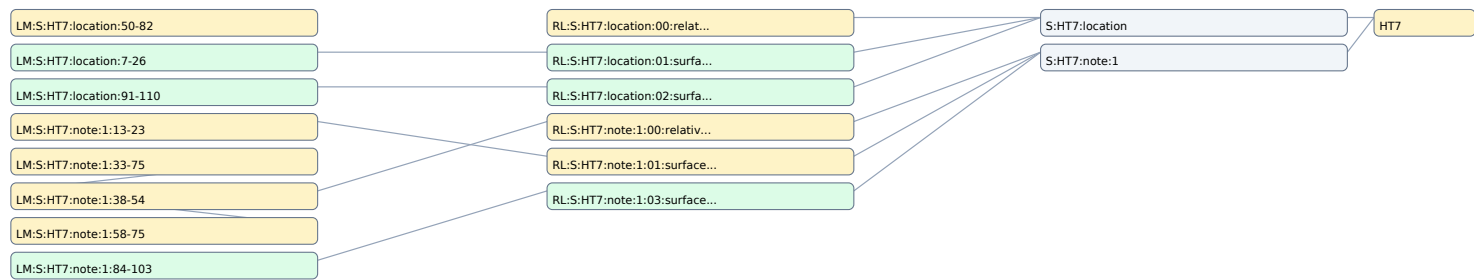
RL:S:HT7:location:02:surface-landmark | op=surface_membership | semantic=['LM:S:HT7:location:91-110'] | executable=['LM:S:HT7:location:91-110'] | dir=None | branch=None | source=S:HT7:location | semantic_hash=6aff53799bb1be4f204ae9b6e66d34ccfbfdad2204aa065d582bfadfe1e49a | status=RESOLVED | reason=None

RL:S:HT7:note:1:00:relative-to | op=directional_relation | semantic=['LM:S:HT7:note:1:33-75'] | executable=['LM:S:HT7:note:1:38-54'] | dir=radial | branch=None | source=S:HT7:note:1 | semantic_hash=daa8e5778bc3a234af22e04dc03d4d73f567d2b62ff04cc1b6971f7eeb851941 | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:HT7:note:1:01:surface-landmark | op=surface_membership | semantic=['LM:S:HT7:note:1:13-23'] | executable=['LM:S:HT7:note:1:13-23'] | dir=None | branch=None | source=S:HT7:note:1 | semantic_hash=bee8b90084b7d9bea0b325b2514ef4f5be8438eaab34c570b80b6bb672aea668 | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:HT7:note:1:03:surface-landmark | op=surface_membership | semantic=['LM:S:HT7:note:1:84-103'] | executable=['LM:S:HT7:note:1:84-103'] | dir=None | branch=None | source=S:HT7:note:1 | semantic_hash=f121f9b7f56f69349e1eb5c8e087590c02097d025a47f1cf7a48115ccbd1b66c | status=RESOLVED | reason=None

Dependency graph



Dependency graph edge list

- RL:S:HT7:location:00:relative-to -> S:HT7:location [relation_to_statement]
- LM:S:HT7:location:7-26 -> RL:S:HT7:location:01:surface-landmark [executable_to_relation]
- RL:S:HT7:location:01:surface-landmark -> S:HT7:location [relation_to_statement]
- LM:S:HT7:location:91-110 -> RL:S:HT7:location:02:surface-landmark [executable_to_relation]
- RL:S:HT7:location:02:surface-landmark -> S:HT7:location [relation_to_statement]
- LM:S:HT7:note:1:33-75 -> LM:S:HT7:note:1:38-54 [semantic_to_executable]
- LM:S:HT7:note:1:38-54 -> RL:S:HT7:note:1:00:relative-to [executable_to_relation]
- RL:S:HT7:note:1:00:relative-to -> S:HT7:note:1 [relation_to_statement]
- LM:S:HT7:note:1:13-23 -> RL:S:HT7:note:1:01:surface-landmark [executable_to_relation]
- RL:S:HT7:note:1:01:surface-landmark -> S:HT7:note:1 [relation_to_statement]
- LM:S:HT7:note:1:84-103 -> RL:S:HT7:note:1:03:surface-landmark [executable_to_relation]
- RL:S:HT7:note:1:03:surface-landmark -> S:HT7:note:1 [relation_to_statement]
- LM:S:HT7:note:1:58-75 -> LM:S:HT7:note:1:38-54 [parent_to_child_subfeature]
- S:HT7:location -> HT7 [statement_to_point]
- S:HT7:note:1 -> HT7 [statement_to_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature
- [] distinct subfeatures
- [] relation operator
- [] condition state
- [] RESOLVED/UNRESOLVED
- [] ACCEPT
- [] REJECT

LI4: RESOLVED -> RESOLVED

Defect family: parent_only_composite_resolution, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, direction_preservation_failure, relation_semantic_hash_mismatch, operand_trace_incomplete, parent_only_composite_resolution, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete

WHO source

location | S:LI4:location | page 44 | On the dorsum of the hand, radial to the midpoint of the second metacarpal bone.

Semantic decomposition

LM:S:LI4:location:36-79 | source= the midpoint of the second metacarpal bone | class=bone.bone | parent=None | child=None | executor=derived_relation_output | status=RESOLVED | identity={"concept_id":null,"constructed_id":"DERIVED:RL:S:LI4:location:01:midpoint-of-entity:b6b15e7a270e7a6c","construction_rule":"entity_midpoint","derived_geometry_id":"DERIVED:RL:S:LI4:location:01:midpoint-of-entity:b6b15e7a270e7a6c","fma_id":null,"geometry_hash":"b6b15e7a270e7a6ceb3bac4be24f32fe29dabc363cff491dcafc4dcdb778d7d5","kind":"derived_relation_output","parent_landmark_id":null,"part_id":null,"producer_relation_id":"RL:S:LI4:location:01:midpoint-of-entity","reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:LI4:location:53-79 | source=the second metacarpal bone | class=bone.bone | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA23900","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:LI4:location:7-25 | source=dorsum of the hand | class=surface.aspect | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"660965ac347a4b5e19c1f02c3d45a8be59be919076604c2a490ce1a7242bf5f4","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:dorsum_hand","subfeature_type":null} | reason=None

Composite binding view

Relation execution

RL:S:LI4:location:01:midpoint-of-entity | op=entity_midpoint | semantic=['LM:S:LI4:location:53-79'] | executable=['LM:S:LI4:location:53-79'] | dir=None | branch=None | source=S:LI4:location | semantic_hash=c29f99d796ea2257c62fce36fdcd0689b41dbb2fe1de7fb1eb5b2bcb051dfbf | status=RESOLVED | reason=None

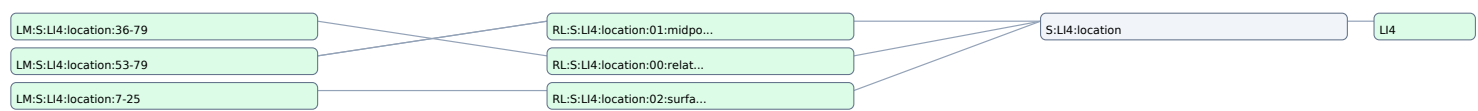
RL:S:LI4:location:00:relative-to | op=directional_relation | semantic=['LM:S:LI4:location:36-79'] | executable=['LM:S:LI4:location:36-79'] | dir=radial | branch=None | source=S:LI4:location | semantic_hash=729d52b919e866632cae68e588cdb65be172ba4c1b97ea1fb85372bd8d059be6 | status=RESOLVED | reason=None

RL:S:LI4:location:02:surface-landmark | op=surface_membership | semantic=['LM:S:LI4:location:7-25'] | executable=['LM:S:LI4:location:7-25'] | dir=None | branch=None | source=S:LI4:location | semantic_hash=2f8b3039cbe7114c2fbdc3ba17a920905d5c4524800aff30d84c85f64ceb3ede | status=RESOLVED | reason=None

Derived geometry outputs

{"derived_geometry_id":"DERIVED:RL:S:LI4:location:01:midpoint-of-entity:b6b15e7a270e7a6c","geometry_hash":"b6b15e7a270e7a6ceb3bac4be24f32fe29dabc363cff491dcafc4dcdb778d7d5","operator":"entity_midpoint","producer_relation_id":"RL:S:LI4:location:01:midpoint-of-entity","semantic_argument_node_ids":["LM:S:LI4:location:53-79"],"source_statement_id":"S:LI4:location"}

Dependency graph



Dependency graph edge list

LM:S:LI4:location:53-79 -> RL:S:LI4:location:01:midpoint-of-entity [executable_to_relation]
RL:S:LI4:location:01:midpoint-of-entity -> S:LI4:location [relation_to_statement]
LM:S:LI4:location:36-79 -> RL:S:LI4:location:00:relative-to [executable_to_relation]
RL:S:LI4:location:00:relative-to -> S:LI4:location [relation_to_statement]
LM:S:LI4:location:7-25 -> RL:S:LI4:location:02:surface-landmark [executable_to_relation]
RL:S:LI4:location:02:surface-landmark -> S:LI4:location [relation_to_statement]
LM:S:LI4:location:53-79 -> RL:S:LI4:location:01:midpoint-of-entity [operand_to_derived_operator]
RL:S:LI4:location:01:midpoint-of-entity -> DERIVED:RL:S:LI4:location:01:midpoint-of-entity:b6b15e7a270e7a6c [operator_to_derived_geometry]
DERIVED:RL:S:LI4:location:01:midpoint-of-entity:b6b15e7a270e7a6c -> LM:S:LI4:location:36-79 [derived_geometry_to_semantic_operand]
S:LI4:location -> LI4 [statement_to_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature
- [] distinct subfeatures
- [] relation operator
- [] condition state

[] RESOLVED/UNRESOLVED □□□ □□□
[] ACCEPT
[] REJECT

ST1: UNRESOLVED -> UNRESOLVED

Defect family: relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, direction_preservation_failure, relation_semantic_hash_mismatch, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete

WHO source

location | S:ST1:location | page 55 | On the face, between the eyeball and the infraorbital margin, directly inferior to the pupil.

Semantic decomposition

LM:S:ST1:location:25-32 | source=eyeball | class=soft_tissue_feature | parent=None | child=None | executor=atlas_name_resolution | status=MULTIPLE | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=multiple atlas-name candidates

LM:S:ST1:location:37-60 | source=the infraorbital margin | class=boundary.border | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:ST1:location:7-11 | source=face | class=region.body_region | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA24728","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:ST1:location:82-92 | source= the pupil | class=soft_tissue_feature | parent=None | child=None | executor=no_family_executor | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=unsupported or unresolved landmark: geometry_or_surface/soft_tissue_feature

Composite binding view

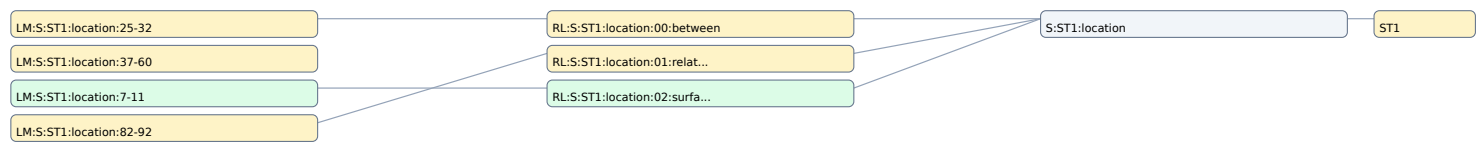
Relation execution

RL:S:ST1:location:00:between | op=between_constraint | semantic=['LM:S:ST1:location:25-32', 'LM:S:ST1:location:37-60'] | executable=['LM:S:ST1:location:25-32'] | dir=None | branch=None | source=S:ST1:location | semantic_hash=945e61339b4d959c1157a90a2c685f3955fa4253452a0ce400113a6568650009 | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:ST1:location:01:relative-to | op=directional_relation | semantic=['LM:S:ST1:location:82-92'] | executable=['LM:S:ST1:location:82-92'] | dir=inferior | branch=None | source=S:ST1:location | semantic_hash=72db92d2f2968b996e40878b1ad74553cec4753e9ccaffad48300667118d81c4 | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:ST1:location:02:surface-landmark | op=surface_membership | semantic=['LM:S:ST1:location:7-11'] | executable=['LM:S:ST1:location:7-11'] | dir=None | branch=None | source=S:ST1:location | semantic_hash=8ccfa9894fa6966b554ed56635305bcb8ad1e606d41e5cc1e89da1bb9cf02a7 | status=RESOLVED | reason=None

Dependency graph



Dependency graph edge list

LM:S:ST1:location:25-32 -> RL:S:ST1:location:00:between [executable_to_relation]
RL:S:ST1:location:00:between -> S:ST1:location [relation_to_statement]
LM:S:ST1:location:82-92 -> RL:S:ST1:location:01:relative-to [executable_to_relation]
RL:S:ST1:location:01:relative-to -> S:ST1:location [relation_to_statement]
LM:S:ST1:location:7-11 -> RL:S:ST1:location:02:surface-landmark [executable_to_relation]
RL:S:ST1:location:02:surface-landmark -> S:ST1:location [relation_to_statement]
S:ST1:location -> ST1 [statement_to_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature execution
- [] distinct subfeatures distinct executable identity
- [] relation operator operand
- [] condition state
- [] RESOLVED/UNRESOLVED
- [] ACCEPT
- [] REJECT

GB14: UNRESOLVED -> UNRESOLVED

Defect family: relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, direction_preservation_failure, quantitative_relation_semantics_loss, quantitative_relation_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, direction_preservation_failure, quantitative_relation_semantics_loss, quantitative_relation_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete

WHO source

location | S:GB14:location | page 187 | On the head, 1 B-cun superior to the eye-brow, directly superior to the centre of the pupil.

Semantic decomposition

LM:S:GB14:location:32-45 | source= the eye-brow | class=soft_tissue_feature | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":FMA54237,"geometry_hash":null,"kind":fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:GB14:location:67-91 | source= the centre of the pupil | class=soft_tissue_feature | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:GB14:location:7-11 | source=head | class=region.body_region | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":FMA7154,"geometry_hash":null,"kind":fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:GB14:location:72-78 | source=centre | class=None | parent=LM:S:GB14:location:82-91 | child=centre | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=parent entity is not uniquely resolved

LM:S:GB14:location:82-91 | source=the pupil | class=soft_tissue_feature | parent=None | child=None | executor=no_family_executor | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=unsupported or unresolved landmark: geometry_or_surface/soft_tissue_feature

Composite binding view

PARENT LM:S:GB14:location:82-91 the pupil status=UNRESOLVED identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:GB14:location:72-78 centre status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

Relation execution

RL:S:GB14:location:02:center-of | op=entity_center | semantic=[LM:S:GB14:location:82-91'] | executable=[LM:S:GB14:location:82-91'] | dir=None | branch=None | source=S:GB14:location | semantic_hash=6085b5e92aa68aba73245695548534f21c3f7fc24625ab3feea0c2cdcb2aa0a9 | status=UNRESOLVED | reason=derived geometry producer operand not executable

RL:S:GB14:location:00:relative-to | op=directional_relation | semantic=[LM:S:GB14:location:32-45'] | executable=[LM:S:GB14:location:32-45'] | dir=superior | branch=None | source=S:GB14:location | semantic_hash=31aeb8150c7b1f4925fee57a91950886412b827952d48e8886960575d2a33ca6 | status=RESOLVED | reason=None

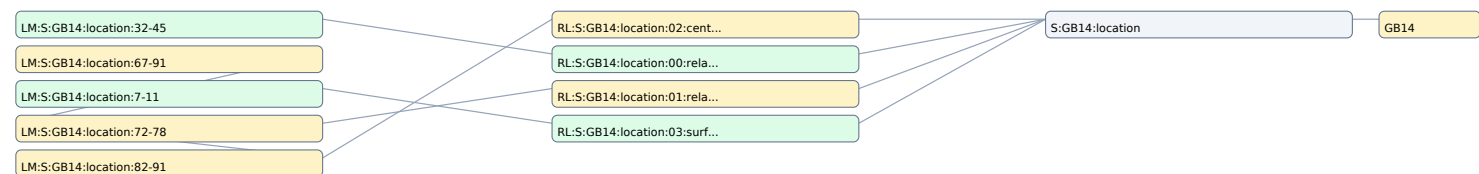
RL:S:GB14:location:01:relative-to | op=directional_relation | semantic=[LM:S:GB14:location:67-91'] | executable=[LM:S:GB14:location:72-78'] | dir=superior | branch=None | source=S:GB14:location | semantic_hash=ee8657682989e68528998b9cb90d0d95d081bb3ea927d2272ab7f893acb62d6c | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:GB14:location:03:surface-landmark | op=surface_membership | semantic=[LM:S:GB14:location:7-11'] | executable=[LM:S:GB14:location:7-11'] | dir=None | branch=None | source=S:GB14:location | semantic_hash=e3ec049bccb909e704731a60d88f5a4cc407c1de9f502f86a374edc203eb92a7 | status=RESOLVED | reason=None

Quantitative measurement semantics

{"anchor_landmark_id":LM:S:GB14:location:32-45,"direction":superior,"executor":approved_calibration,"measurement_id":PM:S:GB14:location:00,"reason":null,"source_statement_id":S:GB14:location,"source_unit":B-cun,"status":RESOLVED,"unit":B-cun,"value":1.0,"who_pdf_page":187}

Dependency graph



Dependency graph edge list

LM:S:GB14:location:82-91 -> RL:S:GB14:location:02:center-of [executable_to_relation]

RL:S:GB14:location:02:center-of -> S:GB14:location [relation_to_statement]
LM:S:GB14:location:32-45 -> RL:S:GB14:location:00:relative-to [executable_to_relation]
PM:S:GB14:location:00 -> RL:S:GB14:location:00:relative-to [measurement_to_relation]
RL:S:GB14:location:00:relative-to -> S:GB14:location [relation_to_statement]
LM:S:GB14:location:67-91 -> LM:S:GB14:location:72-78 [semantic_to_executable]
LM:S:GB14:location:72-78 -> RL:S:GB14:location:01:relative-to [executable_to_relation]
PM:S:GB14:location:00 -> RL:S:GB14:location:01:relative-to [measurement_to_relation]
RL:S:GB14:location:01:relative-to -> S:GB14:location [relation_to_statement]
LM:S:GB14:location:7-11 -> RL:S:GB14:location:03:surface-landmark [executable_to_relation]
RL:S:GB14:location:03:surface-landmark -> S:GB14:location [relation_to_statement]
LM:S:GB14:location:82-91 -> LM:S:GB14:location:72-78 [parent_to_child_subfeature]
LM:S:GB14:location:32-45 -> PM:S:GB14:location:00 [anchor_to_measurement]
PM:S:GB14:location:00 -> S:GB14:location [measurement_to_statement]
S:GB14:location -> GB14 [statement_to_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature
- [] distinct subfeatures
- [] relation operator
- [] condition state
- [] RESOLVED/UNRESOLVED
- [] ACCEPT
- [] REJECT

GB23: RESOLVED -> RESOLVED

Defect family: relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, direction_preservation_failure, quantitative_relation_semantics_loss, quantitative_relation_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete

WHO source

location | S:GB23:location | page 192 | In the lateral thoracic region, in the fourth intercostal space, 1 B-cun anterior to the midaxillary line.

Semantic decomposition

LM:S:GB23:location:39-63 | source=fourth intercostal space | class=space.anatomical_space | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"c8e8691e4f765638eb499518792d13bea202f0a5aa3a1f90a050c51fb2e2ed36","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:fourth_intercostal_space","subfeature_type":null} | reason=None

LM:S:GB23:location:7-30 | source=lateral thoracic region | class=region.body_region | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"8cf739f6e831150ec0c56595886dd8d46bc412a09ff8f132ad1ae55e653d9da","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:lateral_thorax","subfeature_type":null} | reason=None

LM:S:GB23:location:84-105 | source= the midaxillary line | class=line.anatomical_line | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"643f43e0388878b8eb9f6175a1c685aeb1b7b80df0274d8adc5b0b25cfda1420","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:midaxillary_line","subfeature_type":null} | reason=None

Composite binding view

Relation execution

RL:S:GB23:location:00:relative-to | op=directional_relation | semantic=['LM:S:GB23:location:84-105'] | executable=['LM:S:GB23:location:84-105'] | dir=anterior | branch=None | source=S:GB23:location | semantic_hash=cd5fa6b386e8e18634394cbc649be447811a98a48700266100a5bb46bf606a35 | status=RESOLVED | reason=None

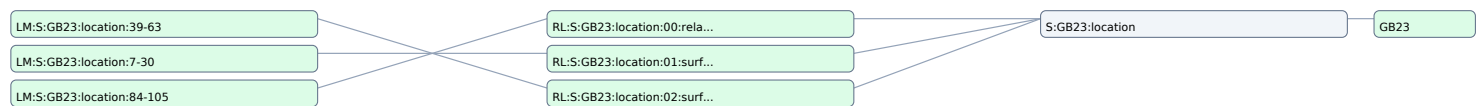
RL:S:GB23:location:01:surface-landmark | op=surface_membership | semantic=['LM:S:GB23:location:7-30'] | executable=['LM:S:GB23:location:7-30'] | dir=None | branch=None | source=S:GB23:location | semantic_hash=70edc62aa335f7f8b5493dcf30455c96e68058cc9ff713660f5cc7e5cb84727b | status=RESOLVED | reason=None

RL:S:GB23:location:02:surface-landmark | op=surface_membership | semantic=['LM:S:GB23:location:39-63'] | executable=['LM:S:GB23:location:39-63'] | dir=None | branch=None | source=S:GB23:location | semantic_hash=1bfe69a87b8c2dd5884e1fcf32ad552fc58d27c4347358a9a53f3b83c26e3838 | status=RESOLVED | reason=None

Quantitative measurement semantics

{ "anchor_landmark_id": "LM:S:GB23:location:84-105", "direction": "anterior", "executor": "approved_calibration", "measurement_id": "PM:S:GB23:location:00", "reason": null, "source_statement_id": "S:GB23:location", "source_unit": "B-cun", "status": "RESOLVED", "unit": "B-cun", "value": 1.0, "who_pdf_page": 192 }

Dependency graph



Dependency graph edge list

- LM:S:GB23:location:84-105 -> RL:S:GB23:location:00:relative-to [executable_to_relation]
- PM:S:GB23:location:00 -> RL:S:GB23:location:00:relative-to [measurement_to_relation]
- RL:S:GB23:location:00:relative-to -> S:GB23:location [relation_to_statement]
- LM:S:GB23:location:7-30 -> RL:S:GB23:location:01:surface-landmark [executable_to_relation]
- RL:S:GB23:location:01:surface-landmark -> S:GB23:location [relation_to_statement]
- LM:S:GB23:location:39-63 -> RL:S:GB23:location:02:surface-landmark [executable_to_relation]
- RL:S:GB23:location:02:surface-landmark -> S:GB23:location [relation_to_statement]
- LM:S:GB23:location:84-105 -> PM:S:GB23:location:00 [anchor_to_measurement]
- PM:S:GB23:location:00 -> S:GB23:location [measurement_to_statement]
- S:GB23:location -> GB23 [statement_to_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature
- [] distinct subfeatures
- [] distinct subfeatures
- [] distinct executable identity
- [] relation operator
- [] operand

[] condition state
[] RESOLVED/UNRESOLVED
[] ACCEPT
[] REJECT

LU6: UNRESOLVED -> UNRESOLVED

[illegible]

WHO source

location | S:LU6:location | page 37 | On the anterolateral aspect of the forearm, on the line connecting LU5 with LU9, 7 B-cun superior to the palmar wrist crease.

note | S:LU6:note:1 | page 37 | Note: LU6 is 5 B-cun inferior to LU5, 1 B-cun superior to the midpoint of the line connecting LU5 with LU9.

Semantic decomposition

LM:S:LU6:location:100-124 | source= the palmar wrist crease | class=crease.skin_crease | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"383e200f286d70d3c13689830673e15bb470bf299882294478ffe17ce465fa2f","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:palmar_wrist_crease","subfeature_type":null} | reason=None

```
LM:S:LU6:location:51-66 | source=line connecting | class=line.reference_line | parent=None | child=None | executor=no_family_executor |
status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":nul
l,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} |
reason=unsupported or unresolved landmark: geometry or surface/line reference line
```

```
LM:S:LU6:location:67-70 | source=LU5 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED |
identity="{concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"reference_acu
point":{"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}} | reason=None
```

LM:S:LU6:location:7-27 | source=anterolateral aspect | class=surface.aspect | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"8ce43d41eaba7ffc905ab399e621546e96abede4af22e20655f5814a9f8b665c","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point":null,"registry_id":"SR:anterolateral forearm","subfeature_type":null} | reason=None

```
LM:S:LU6:location:76-79 | source=LU9 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED |
identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"reference_acu
point":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None
```

LM:S:LU6:note:1:103-106 | source=LU9 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"reference_acupoint":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

```
LM:S:LU6:note:1:32-36 | source= LU5: class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED |
identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"reference_acu
point":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None
```

```
LM:S:LU6:note:1:57-106 | source= the midpoint of the line connecting LU5 with LU9 | class=line.reference_line | parent=None | child=None |
executor=derived_relation_output | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"DERIVED:RL:S:LU6:note:1:01:midpoint-between:679c23
90a3beb02b","construction_rule":"midpoint_between","derived_geometry_id":null,"DERIVED:RL:S:LU6:note:1:01:midpoint-between:679c2390a3beb02b","fma_id":n
ull,"geometry_hash":"679c2390a3beb02b22e50e4b4b33fe4be50f4c00b20355042ce14e6d274e1d1c","kind":"derived_relation_output","parent_landmark_id":n
ull,"part_id":null,"producer_relation_id":"RL:S:LU6:note:1:01:midpoint-between","reference_point_id":null,"registry_id":null,"subfeature_type":null} |
reason=None
```

LM:S:LU6:note:1:94-97 | source=LU5 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"LU5","registry_id":null,"subfeature_type":null} | reason=None

LM21:S:LU6:location:67-70:002c0a95e5 | source=LU5 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"LU5","registry_id":null,"subfeature_type":null} | reason=None

LM21:S:LU6:location:76-79:30c4b43ed7 | source=LU9 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"LU9","registry_id":null,"subfeature_type":null} | reason=None

Composite binding view

Relation execution

RL:S:LU6:note:1:01:midpoint-between | op=midpoint_between | semantic=['LM:S:LU6:note:1:94-97', 'LM:S:LU6:note:1:103-106'] | executable=['LM:S:LU6:note:1:94-97', 'LM:S:LU6:note:1:103-106'] | dir=None | branch=None | source=S:LU6:note:1 | semantic_hash=887c0c383f13081f7d5fc9dcb7d24623b6384c9169864b732a40ae7a1e2459a9 | status=RESOLVED | reason=None

RL:S:LU6:location:00:on-line | op=line_membership | semantic=['LM:S:LU6:location:67-70', 'LM:S:LU6:location:76-79'] | executable=['LM:S:LU6:location:67-70', 'LM:S:LU6:location:76-79'] | dir=None | branch=None | source=S:LU6:location | semantic_hash=c149632278ffa2062a2d6417eda595794a494728900361ca50ec94294ce07339 | status=RESOLVED | reason=None

RL:S:LU6:location:01:relative-to | op=directional_relation | semantic=['LM:S:LU6:location:100-124'] | executable=['LM:S:LU6:location:100-124'] | dir=superior | branch=None | source=S:LU6:location | semantic_hash=7d1b876143780323224bcb22df09847a84b8c03711c20722aff58b898f5e9e34 | status=RESOLVED | reason=None

RL:S:LU6:location:02:surface-landmark | op=surface_membership | semantic=['LM:S:LU6:location:7-27'] | executable=['LM:S:LU6:location:7-27'] | dir=None | branch=None | source=S:LU6:location | semantic_hash=cdd26d7ebbe671e12c116b46f31acfdc8cea23fa7503ff9d2e77df9e9dc229b | status=RESOLVED | reason=None

RL:S:LU6:location:03:surface-landmark | op=surface_membership | semantic=['LM:S:LU6:location:51-66'] | executable=['LM:S:LU6:location:51-66'] | dir=None | branch=None | source=S:LU6:location | semantic_hash=8e04ef526e23db75daeac925968df1f8c5b83f7998668c4b3f242b24f32284f5 | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:LU6:note:1:00:on-line | op=line_membership | semantic=['LM:S:LU6:note:1:94-97', 'LM:S:LU6:note:1:103-106'] | executable=['LM:S:LU6:note:1:94-97', 'LM:S:LU6:note:1:103-106'] | dir=None | branch=None | source=S:LU6:note:1 | semantic_hash=4f315fd058c9a7fa7d2d94f7678ae905f05d2bf5f8a69f6d9c9bd22d222141103 | status=RESOLVED | reason=None

RL:S:LU6:note:1:02:relative-to | op=directional_relation | semantic=['LM:S:LU6:note:1:32-36'] | executable=['LM:S:LU6:note:1:32-36'] | dir=inferior | branch=None | source=S:LU6:note:1 | semantic_hash=74fcb6792f0bb6b842c265808f002e755755949e6886dfed6b2e4b2854e0c2754 | status=RESOLVED | reason=None

RL:S:LU6:note:1:03:relative-to | op=directional_relation | semantic=['LM:S:LU6:note:1:57-106'] | executable=['LM:S:LU6:note:1:57-106'] | dir=superior | branch=None | source=S:LU6:note:1 | semantic_hash=5894bcf1c807792b9e71f9897df5ad86c3c983fb221121c98fadbb256ee2e9 | status=RESOLVED | reason=None

RL21:S:LU6:location:67-70:reference-acupoint:f42e2db8d9 | op=reference_dependency | semantic=['LM21:S:LU6:location:67-70:002c0a95e5'] | executable=['LM21:S:LU6:location:67-70:002c0a95e5'] | dir=None | branch=None | source=S:LU6:location | semantic_hash=05f1807669ca1fde29550c56cfdc9644e5ac7ceb1bad4c5bd02d5b9e84b34f5a | status=RESOLVED | reason=None

RL21:S:LU6:location:76-79:reference-acupoint:5171900d45 | op=reference_dependency | semantic=['LM21:S:LU6:location:76-79:30c4b43ed7'] | executable=['LM21:S:LU6:location:76-79:30c4b43ed7'] | dir=None | branch=None | source=S:LU6:location | semantic_hash=029a20d19605214d23599a39cce266395129802c1f79bf6e015b25f728db806e | status=RESOLVED | reason=None

RL21:S:LU6:note:1:33-36:reference-acupoint:a79a726278 | op=reference_dependency | semantic=['LM:S:LU6:note:1:32-36'] | executable=['LM:S:LU6:note:1:32-36'] | dir=None | branch=None | source=S:LU6:note:1 | semantic_hash=5a5d921b81fcfb3f34f697f85f63bbf4476ce89154f942f0c2218a0dc3dabb | status=RESOLVED | reason=None

RL21:S:LU6:note:1:94-97:reference-acupoint:0d3fedaf0b | op=reference_dependency | semantic=['LM:S:LU6:note:1:94-97'] | executable=['LM:S:LU6:note:1:94-97'] | dir=None | branch=None | source=S:LU6:note:1 | semantic_hash=28577bdaac0375aef4129c59251a914aadaa8a22e4d79b53f8c5e6622647471c | status=RESOLVED | reason=None

RL21:S:LU6:note:1:103-106:reference-acupoint:ffaf5ca270 | op=reference_dependency | semantic=['LM:S:LU6:note:1:103-106'] | executable=['LM:S:LU6:note:1:103-106'] | dir=None | branch=None | source=S:LU6:note:1 | semantic_hash=b0b202b83a609a46835ccaf99ca7a1d56f2c85a9528bae3c70c5733321b0aa75 | status=RESOLVED | reason=None

Quantitative measurement semantics

{"anchor_landmark_id":"LM:S:LU6:location:100-124","direction":"superior","executor":"approved_calibration","measurement_id":"PM:S:LU6:location:00","reason":null,"source_statement_id":"S:LU6:location","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":7.0,"who_pdf_page":37}

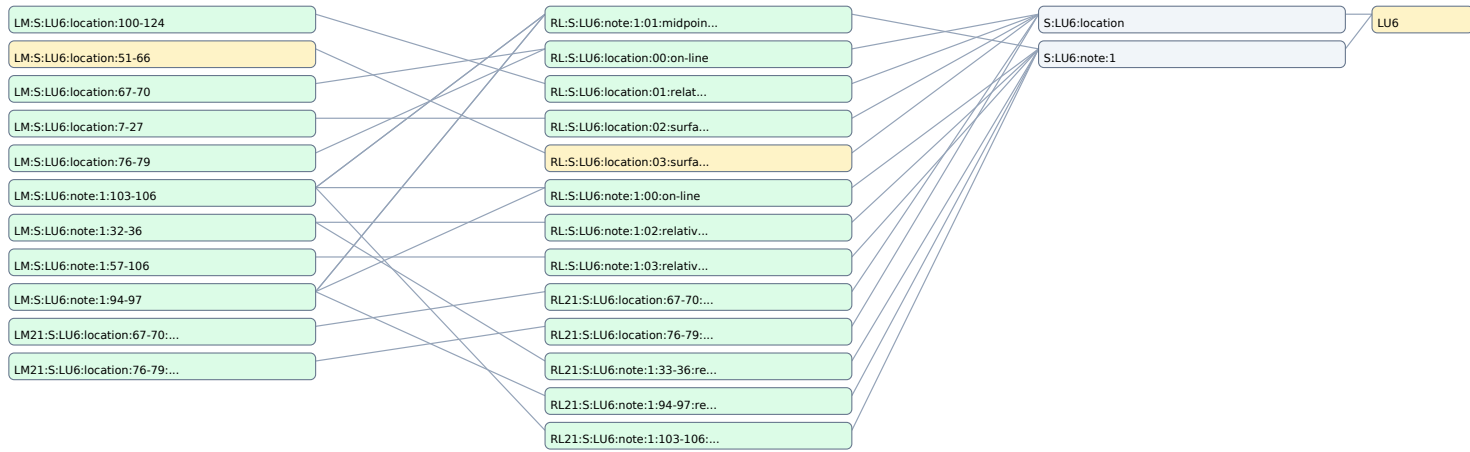
{"anchor_landmark_id":"LM:S:LU6:note:1:32-36","direction":"inferior","executor":"approved_calibration","measurement_id":"PM:S:LU6:note:1:00","reason":null,"source_statement_id":"S:LU6:note:1","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":5.0,"who_pdf_page":37}

{"anchor_landmark_id":"LM:S:LU6:note:1:57-106","direction":"superior","executor":"approved_calibration","measurement_id":"PM:S:LU6:note:1:01","reason":null,"source_statement_id":"S:LU6:note:1","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":1.0,"who_pdf_page":37}

Derived geometry outputs

{"derived_geometry_id":"DERIVED:RL:S:LU6:note:1:01:midpoint-between:679c2390a3beb02b","geometry_hash":"679c2390a3beb02b22e50e4b4b33fe4be50f4c00b20355042ce14e6d274e1d1c","operator":"midpoint_between","producer_relation_id":"RL:S:LU6:note:1:01:midpoint-between","semantic_argument_node_ids":["LM:S:LU6:note:1:94-97","LM:S:LU6:note:1:103-106"],"source_statement_id":"S:LU6:note:1"}

Dependency graph



Dependency graph edge list

LM:S:LU6:note:1:94-97 -> RL:S:LU6:note:1:01:midpoint-between [executable_to_relation]

LM:S:LU6:note:1:103-106 -> RL:S:LU6:note:1:01:midpoint-between [executable_to_relation]

RL:S:LU6:note:1:01:midpoint-between -> S:LU6:note:1 [relation_to_statement]

LM:S:LU6:location:67-70 -> RL:S:LU6:location:00:on-line [executable_to_relation]

LM:S:LU6:location:76-79 -> RL:S:LU6:location:00:on-line [executable_to_relation]

RL:S:LU6:location:00:on-line -> S:LU6:location [relation_to_statement]

LM:S:LU6:location:100-124 -> RL:S:LU6:location:01:relative-to [executable_to_relation]

PM:S:LU6:location:00 -> RL:S:LU6:location:01:relative-to [measurement_to_relation]

RL:S:LU6:location:01:relative-to -> S:LU6:location [relation_to_statement]

LM:S:LU6:location:7-27 -> RL:S:LU6:location:02:surface-landmark [executable_to_relation]

RL:S:LU6:location:02:surface-landmark -> S:LU6:location [relation_to_statement]

LM:S:LU6:location:51-66 -> RL:S:LU6:location:03:surface-landmark [executable_to_relation]

RL:S:LU6:location:03:surface-landmark -> S:LU6:location [relation_to_statement]

LM:S:LU6:note:1:94-97 -> RL:S:LU6:note:1:00:on-line [executable_to_relation]

LM:S:LU6:note:1:103-106 -> RL:S:LU6:note:1:00:on-line [executable_to_relation]

RL:S:LU6:note:1:00:on-line -> S:LU6:note:1 [relation_to_statement]

LM:S:LU6:note:1:32-36 -> RL:S:LU6:note:1:02:relative-to [executable_to_relation]

PM:S:LU6:note:1:00 -> RL:S:LU6:note:1:02:relative-to [measurement_to_relation]

RL:S:LU6:note:1:02:relative-to -> S:LU6:note:1 [relation_to_statement]

LM:S:LU6:note:1:57-106 -> RL:S:LU6:note:1:03:relative-to [executable_to_relation]

PM:S:LU6:note:1:01 -> RL:S:LU6:note:1:03:relative-to [measurement_to_relation]

RL:S:LU6:note:1:03:relative-to -> S:LU6:note:1 [relation_to_statement]

LM21:S:LU6:location:67-70:002c0a95e5 -> RL21:S:LU6:location:67-70:reference-acupoint:f42e2db8d9 [executable_to_relation]

RL21:S:LU6:location:67-70:reference-acupoint:f42e2db8d9 -> S:LU6:location [relation_to_statement]

LM21:S:LU6:location:76-79:30c4b43ed7 -> RL21:S:LU6:location:76-79:reference-acupoint:5171900d45 [executable_to_relation]

RL21:S:LU6:location:76-79:reference-acupoint:5171900d45 -> S:LU6:location [relation_to_statement]

LM:S:LU6:note:1:32-36 -> RL21:S:LU6:note:1:33-36:reference-acupoint:a79a726278 [executable_to_relation]

RL21:S:LU6:note:1:33-36:reference-acupoint:a79a726278 -> S:LU6:note:1 [relation_to_statement]

LM:S:LU6:note:1:94-97 -> RL21:S:LU6:note:1:94-97:reference-acupoint:0d3fedaf0b [executable_to_relation]

RL21:S:LU6:note:1:94-97:reference-acupoint:0d3fedaf0b -> S:LU6:note:1 [relation_to_statement]

LM:S:LU6:note:1:103-106 -> RL21:S:LU6:note:1:103-106:reference-acupoint:ffaf5ca270 [executable_to_relation]

RL21:S:LU6:note:1:103-106:reference-acupoint:ffaf5ca270 -> S:LU6:note:1 [relation_to_statement]

LM:S:LU6:location:100-124 -> PM:S:LU6:location:00 [anchor_to_measurement]

PM:S:LU6:location:00 -> S:LU6:location [measurement_to_statement]

LM:S:LU6:note:1:32-36 -> PM:S:LU6:note:1:00 [anchor_to_measurement]

PM:S:LU6:note:1:00 -> S:LU6:note:1 [measurement_to_statement]

LM:S:LU6:note:1:57-106 -> PM:S:LU6:note:1:01 [anchor_to_measurement]

PM:S:LU6:note:1:01 -> S:LU6:note:1 [measurement_to_statement]

LM:S:LU6:note:1:94-97 -> RL:S:LU6:note:1:01:midpoint-between [operand_to_derived_operator]

LM:S:LU6:note:1:103-106 -> RL:S:LU6:note:1:01:midpoint-between [operand_to_derived_operator]
RL:S:LU6:note:1:01:midpoint-between -> DERIVED:RL:S:LU6:note:1:01:midpoint-between:679c2390a3beb02b [operator_to_derived_geometry]
DERIVED:RL:S:LU6:note:1:01:midpoint-between:679c2390a3beb02b -> LM:S:LU6:note:1:57-106 [derived_geometry_to_semantic_operand]
S:LU6:location -> LU6 [statement_to_point]
S:LU6:note:1 -> LU6 [statement_to_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature execution
- [] distinct subfeatures distinct executable identity
- [] relation operator operand
- [] condition state
- [] RESOLVED/UNRESOLVED
- [] ACCEPT
- [] REJECT

LI7: UNRESOLVED -> UNRESOLVED

Defect family: relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, direction_preservation_failure, quantitative_relation_semantics_loss, quantitative_relation_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete

WHO source

location | S:LI7:location | page 46 | On the posterolateral aspect of the forearm, on the line connecting LI5 with LI11, 5 B-cun superior to the dorsal wrist crease.

Semantic decomposition

LM:S:LI7:location:102-126 | source= the dorsal wrist crease | class=crease.skin_crease | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"0ff534739841ae12b463e9ea5102d7801e5de256871212f7ddeb84a0b0265288","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:dorsal_wrist_crease","subfeature_type":null} | reason=None

LM:S:LI7:location:52-67 | source=line connecting | class=line.reference_line | parent=None | child=None | executor=no_family_executor | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=unsupported or unresolved landmark: geometry_or_surface/line.reference_line

LM:S:LI7:location:68-71 | source=LI5 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"LI5","registry_id":null,"subfeature_type":null} | reason=None

LM:S:LI7:location:7-28 | source=posterolateral aspect | class=surface.aspect | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"4e542fb31cd71e7a148b3607b7424ea18ab74b0b8ddb0210ce3c5abccc2c54d9","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:posterolateral_forearm","subfeature_type":null} | reason=None

LM:S:LI7:location:77-81 | source=LI11 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"LI11","registry_id":null,"subfeature_type":null} | reason=None

LM21:S:LI7:location:68-71:1bff496df7 | source=LI5 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"LI5","registry_id":null,"subfeature_type":null} | reason=None

LM21:S:LI7:location:77-81:245ea01482 | source=LI11 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"LI11","registry_id":null,"subfeature_type":null} | reason=None

Composite binding view

Relation execution

RL:S:LI7:location:00:on-line | op=line_membership | semantic=['LM:S:LI7:location:68-71', 'LM:S:LI7:location:77-81'] | executable=['LM:S:LI7:location:68-71', 'LM:S:LI7:location:77-81'] | dir=None | branch=None | source=S:LI7:location | semantic_hash=ad6194b0e3b11d92dbca84ebf4a43fa7bc807e551b176bd2cc22501b051f97eb | status=RESOLVED | reason=None

RL:S:LI7:location:01:relative-to | op=directional_relation | semantic=['LM:S:LI7:location:102-126'] | executable=['LM:S:LI7:location:102-126'] | dir=superior | branch=None | source=S:LI7:location | semantic_hash=203d246d22fd077c7d3adaa839d7c9b322f1e1f9ba697198553824c99e81cd08 | status=RESOLVED | reason=None

RL:S:LI7:location:02:surface-landmark | op=surface_membership | semantic=['LM:S:LI7:location:7-28'] | executable=['LM:S:LI7:location:7-28'] | dir=None | branch=None | source=S:LI7:location | semantic_hash=645a7786adfb126f9331ad80acc507fd311a4a8f709b6939ec314bde2678defb | status=RESOLVED | reason=None

RL:S:LI7:location:03:surface-landmark | op=surface_membership | semantic=['LM:S:LI7:location:52-67'] | executable=['LM:S:LI7:location:52-67'] | dir=None | branch=None | source=S:LI7:location | semantic_hash=715cfae4f947de2ffa15a8484d641395cc15af69eacef910d3a765dfcdd9b83 | status=UNRESOLVED | reason=executable operand not uniquely executable

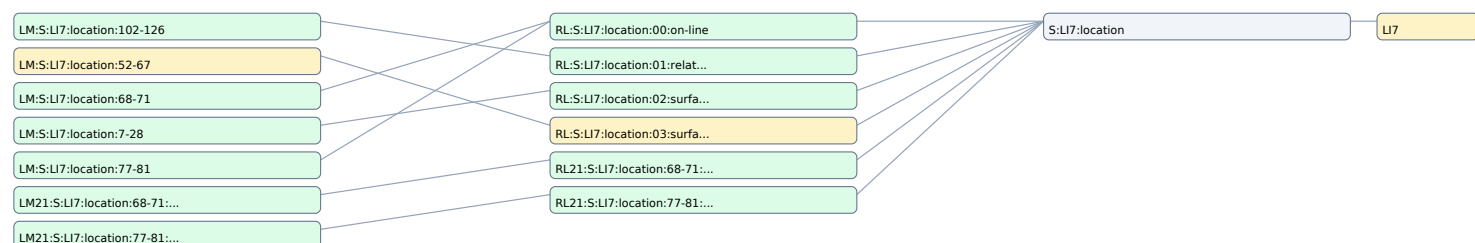
RL21:S:LI7:location:68-71:reference-acupoint:931d5b1cfd | op=reference_dependency | semantic=['LM21:S:LI7:location:68-71:1bff496df7'] | executable=['LM21:S:LI7:location:68-71:1bff496df7'] | dir=None | branch=None | source=S:LI7:location | semantic_hash=ff30b0e4e490dab4f50a663cdb39b1f315a1f4c702ccd535b222bfac060310d5 | status=RESOLVED | reason=None

RL21:S:LI7:location:77-81:reference-acupoint:bbace3ac1c | op=reference_dependency | semantic=['LM21:S:LI7:location:77-81:245ea01482'] | executable=['LM21:S:LI7:location:77-81:245ea01482'] | dir=None | branch=None | source=S:LI7:location | semantic_hash=6b7e1ea62458d1a9ef35a46311e711fe30831478fa95fc4c586b9d8b9a52386f | status=RESOLVED | reason=None

Quantitative measurement semantics

{ "anchor_landmark_id": "LM:S:LI7:location:102-126", "direction": "superior", "executor": "approved_calibration", "measurement_id": "PM:S:LI7:location:00", "reason": null, "source_statement_id": "S:LI7:location", "source_unit": "B-cun", "status": "RESOLVED", "unit": "B-cun", "value": "5.0", "who_pdf_page": 46 }

Dependency graph



Dependency graph edge list

LM:S:LI7:location:68-71 -> RL:S:LI7:location:00:on-line [executable_to_relation]
 LM:S:LI7:location:77-81 -> RL:S:LI7:location:00:on-line [executable_to_relation]
 RL:S:LI7:location:00:on-line -> S:LI7:location [relation_to_statement]
 LM:S:LI7:location:102-126 -> RL:S:LI7:location:01:relative-to [executable_to_relation]
 PM:S:LI7:location:00 -> RL:S:LI7:location:01:relative-to [measurement_to_relation]
 RL:S:LI7:location:01:relative-to -> S:LI7:location [relation_to_statement]
 LM:S:LI7:location:7-28 -> RL:S:LI7:location:02:surface-landmark [executable_to_relation]
 RL:S:LI7:location:02:surface-landmark -> S:LI7:location [relation_to_statement]
 LM:S:LI7:location:52-67 -> RL:S:LI7:location:03:surface-landmark [executable_to_relation]
 RL:S:LI7:location:03:surface-landmark -> S:LI7:location [relation_to_statement]
 LM21:S:LI7:location:68-71:1bff496df7 -> RL21:S:LI7:location:68-71:reference-acupoint:931d5b1cfd [executable_to_relation]
 RL21:S:LI7:location:68-71:reference-acupoint:931d5b1cfd -> S:LI7:location [relation_to_statement]
 LM21:S:LI7:location:77-81:245ea01482 -> RL21:S:LI7:location:77-81:reference-acupoint:bbace3ac1c [executable_to_relation]
 RL21:S:LI7:location:77-81:reference-acupoint:bbace3ac1c -> S:LI7:location [relation_to_statement]
 LM:S:LI7:location:102-126 -> PM:S:LI7:location:00 [anchor_to_measurement]
 PM:S:LI7:location:00 -> S:LI7:location [measurement_to_statement]
 S:LI7:location -> LI7 [statement_to_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

[] WHO source phrase 0000 0000
 [] semantic granularity 0000
 [] child subfeature 000 execution 0000
 [] distinct subfeatures 000 distinct executable identity 0000
 [] relation operator 0000 operand 0000
 [] condition state 0000 0000
 [] RESOLVED/UNRESOLVED 0000 0000
 [] ACCEPT
 [] REJECT

GB26: UNRESOLVED -> UNRESOLVED

Defect family: relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, direction_preservation_failure, relation_semantic_hash_mismatch, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, direction_preservation_failure, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete

WHO source

location | S:GB26:location | page 193 | On the lateral abdomen, inferior to the free extremity of the 11th rib, at the same level as the centre of umbilicus.

note | S:GB26:note:1 | page 193 | Note 1: GB26 can be located by first locating the tenth rib, then searching for the free extremity of the 11th rib that is situated immediately below the inferior border of the costal arch.

note | S:GB26:note:2 | page 193 | Note 2: GB26 is inferior to LR13, at the same level as CV8.

Semantic decomposition

LM:S:GB26:location:107-116 | source=umbilicus | class=orifice_or_cavity | parent=None | child=None | executor=atlas_name_resolution | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=no unique atlas name match

LM:S:GB26:location:35-70 | source= the free extremity of the 11th rib | class=structure.part | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:GB26:location:40-54 | source=free extremity | class=structure.part | parent=LM:S:GB26:location:58-70 | child=free extremity | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=no approved executable construction rule for canonical subfeature 'free_end'

LM:S:GB26:location:58-70 | source=the 11th rib | class=bone.bone | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA8499","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:GB26:location:7-22 | source=lateral abdomen | class=region.body_region | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"912678a19ed1d94961027afd11cd686ebc1dd7d67fe77a92a26935f99509145","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:lateral_abdomen","subfeature_type":null} | reason=None

LM:S:GB26:location:93-116 | source=the centre of umbilicus | class=orifice_or_cavity | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:GB26:location:97-103 | source=centre | class=None | parent=LM:S:GB26:location:107-116 | child=centre | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=parent entity is not uniquely resolved

LM:S:GB26:note:1:103-189 | source=the 11th rib that is situated immediately below the inferior border of the costal arch | class=boundary.border | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:GB26:note:1:85-99 | source=free extremity | class=structure.part | parent=LM:S:GB26:note:1:103-189 | child=free extremity | executor=composite_subfeature_binding | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=parent entity is not uniquely resolved

LM:S:GB26:note:2:27-32 | source= LR13 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"LR13","registry_id":null,"subfeature_type":null} | reason=None

LM:S:GB26:note:2:55-58 | source=CV8 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":"CV8","registry_id":null,"subfeature_type":null} | reason=None

Composite binding view

PARENT LM:S:GB26:location:58-70 the 11th rib status=RESOLVED identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA8499","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:GB26:location:40-54 free extremity status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"

registry_id":null,"subfeature_type":null}

PARENT LM:S:GB26:location:107-116 umbilicus status=UNRESOLVED identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:GB26:location:97-103 centre status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

PARENT LM:S:GB26:note:1:103-189 the 11th rib that is situated immediately below the inferior border of the costal arch status=UNRESOLVED identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

-> CHILD LM:S:GB26:note:1:85-99 free extremity status=UNRESOLVED executable={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null}

Relation execution

RL:S:GB26:location:02:center-of | op=entity_center | semantic=['LM:S:GB26:location:107-116'] | executable=['LM:S:GB26:location:107-116'] | dir=None | branch=None | source=S:GB26:location | semantic_hash=9750aafd0d6a96a6e5319cb643c47bd93bdd699ffcfb1ec8943f1a7764f4b93 | status=UNRESOLVED | reason=derived geometry producer operand not executable

RL:S:GB26:location:00:relative-to | op=directional_relation | semantic=['LM:S:GB26:location:35-70'] | executable=['LM:S:GB26:location:40-54'] | dir=inferior | branch=None | source=S:GB26:location | semantic_hash=3cca02d971ab3af39c419f267949ebc29549b24e9dad210dcd168fd53c88110 | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:GB26:location:01:same-level | op=same_level_plane | semantic=['LM:S:GB26:location:93-116'] | executable=['LM:S:GB26:location:97-103'] | dir=None | branch=None | source=S:GB26:location | semantic_hash=1d67c45945a25af43a30c3ce90df938a21b17353ff0f1dce5c6956444aeaf76e | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:GB26:location:03:surface-landmark | op=surface_membership | semantic=['LM:S:GB26:location:7-22'] | executable=['LM:S:GB26:location:7-22'] | dir=None | branch=None | source=S:GB26:location | semantic_hash=de0cadf3ad87403b3c3b4e22c1b7d4e222ad9e64de193a01fd5caf05f646b180 | status=RESOLVED | reason=None

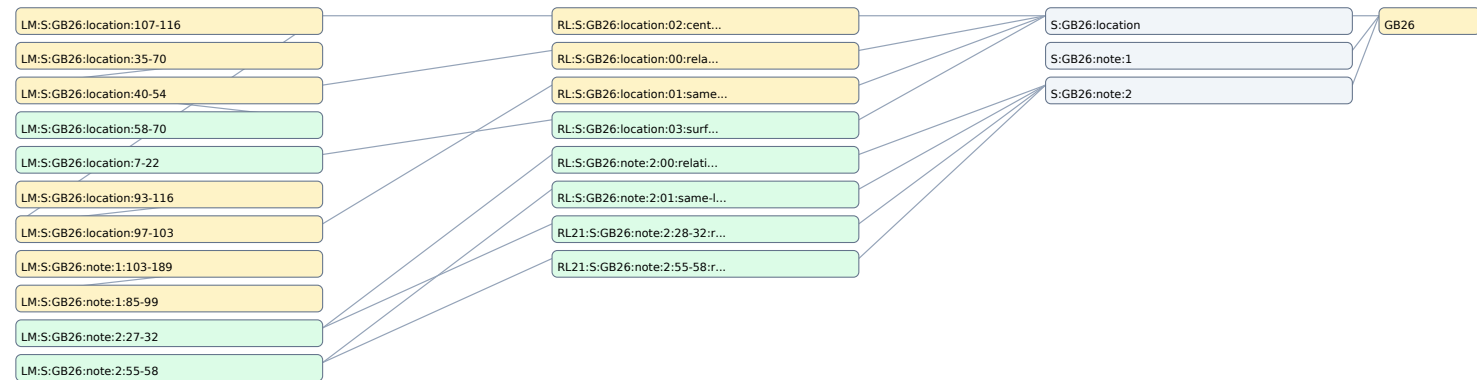
RL:S:GB26:note:2:00:relative-to | op=directional_relation | semantic=['LM:S:GB26:note:2:27-32'] | executable=['LM:S:GB26:note:2:27-32'] | dir=inferior | branch=None | source=S:GB26:note:2 | semantic_hash=05946e573fdf4460558d857860f3bad13eeee1064b15c119d4263b43a77713f4c | status=RESOLVED | reason=None

RL:S:GB26:note:2:01:same-level | op=same_level_plane | semantic=['LM:S:GB26:note:2:55-58'] | executable=['LM:S:GB26:note:2:55-58'] | dir=None | branch=None | source=S:GB26:note:2 | semantic_hash=b349b94b3165d673998a7de014e75c2e796523cdf887a895ccbf9425aa0a7626 | status=RESOLVED | reason=None

RL21:S:GB26:note:2:28-32:reference-acupoint:260f7e9589 | op=reference_dependency | semantic=['LM:S:GB26:note:2:27-32'] | executable=['LM:S:GB26:note:2:27-32'] | dir=None | branch=None | source=S:GB26:note:2 | semantic_hash=64c05c9bbc10431a170f57bf4d38b65c7f7ab0b475213028f9d67133624ea193 | status=RESOLVED | reason=None

RL21:S:GB26:note:2:55-58:reference-acupoint:3e3fd2558d | op=reference_dependency | semantic=['LM:S:GB26:note:2:55-58'] | executable=['LM:S:GB26:note:2:55-58'] | dir=None | branch=None | source=S:GB26:note:2 | semantic_hash=843b3715655f5935f76faf697373a2b61271d7a80a3e8ec83a15bed8bc51fe21 | status=RESOLVED | reason=None

Dependency graph



Dependency graph edge list

LM:S:GB26:location:107-116 -> RL:S:GB26:location:02:center-of [executable_to_relation]

RL:S:GB26:location:02:center-of -> S:GB26:location [relation_to_statement]

LM:S:GB26:location:35-70 -> LM:S:GB26:location:40-54 [semantic_to_executable]

LM:S:GB26:location:40-54 -> RL:S:GB26:location:00:relative-to [executable_to_relation]

RL:S:GB26:location:00:relative-to -> S:GB26:location [relation_to_statement]

LM:S:GB26:location:93-116 -> LM:S:GB26:location:97-103 [semantic_to_executable]

LM:S:GB26:location:97-103 -> RL:S:GB26:location:01:same-level [executable_to_relation]

RL:S:GB26:location:01:same-level -> S:GB26:location [relation_to_statement]

LM:S:GB26:location:7-22 -> RL:S:GB26:location:03:surface-landmark [executable_to_relation]

RL:S:GB26:location:03:surface-landmark -> S:GB26:location [relation_to_statement]

LM:S:GB26:note:2:27-32 -> RL:S:GB26:note:2:00:relative-to [executable_to_relation]

RL:S:GB26:note:2:00:relative-to -> S:GB26:note:2 [relation_to_statement]
LM:S:GB26:note:2:55-58 -> RL:S:GB26:note:2:01:same-level [executable_to_relation]
RL:S:GB26:note:2:01:same-level -> S:GB26:note:2 [relation_to_statement]
LM:S:GB26:note:2:27-32 -> RL21:S:GB26:note:2:28-32:reference-acupoint:260f7e9589 [executable_to_relation]
RL21:S:GB26:note:2:28-32:reference-acupoint:260f7e9589 -> S:GB26:note:2 [relation_to_statement]
LM:S:GB26:note:2:55-58 -> RL21:S:GB26:note:2:55-58:reference-acupoint:3e3fd2558d [executable_to_relation]
RL21:S:GB26:note:2:55-58:reference-acupoint:3e3fd2558d -> S:GB26:note:2 [relation_to_statement]
LM:S:GB26:location:58-70 -> LM:S:GB26:location:40-54 [parent_to_child_subfeature]
LM:S:GB26:location:107-116 -> LM:S:GB26:location:97-103 [parent_to_child_subfeature]
LM:S:GB26:note:1:103-189 -> LM:S:GB26:note:1:85-99 [parent_to_child_subfeature]
S:GB26:location -> GB26 [statement_to_point]
S:GB26:note:1 -> GB26 [statement_to_point]
S:GB26:note:2 -> GB26 [statement_to_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature
- [] distinct subfeatures
- [] relation operator
- [] condition state
- [] RESOLVED/UNRESOLVED
- [] ACCEPT
- [] REJECT

ST2: UNRESOLVED -> UNRESOLVED

Defect family: relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete

WHO source

location | S:ST2:location | page 55 | On the face, in the infraorbital foramen.

Semantic decomposition

LM:S:ST2:location:20-40 | source=infraorbital foramen | class=bone.opening | parent=None | child=None | executor=atlas_name_resolution | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=no unique atlas name match

LM:S:ST2:location:7-11 | source=face | class=region.body_region | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA24728","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

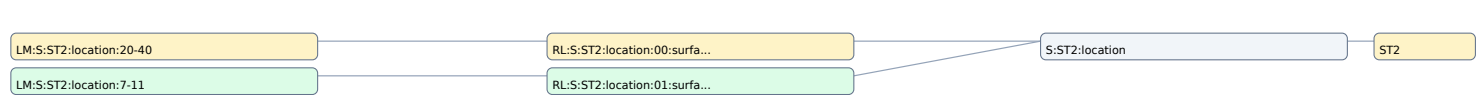
Composite binding view

Relation execution

RL:S:ST2:location:00:surface-landmark | op=surface_membership | semantic=['LM:S:ST2:location:20-40'] | executable=['LM:S:ST2:location:20-40'] | dir=None | branch=None | source=S:ST2:location | semantic_hash=80dfa42923a2d7c22d3091f91df918bd9c1b172b292e2454029e1a00a89a3990 | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:ST2:location:01:surface-landmark | op=surface_membership | semantic=['LM:S:ST2:location:7-11'] | executable=['LM:S:ST2:location:7-11'] | dir=None | branch=None | source=S:ST2:location | semantic_hash=10ecae12ce6e4bde539edd6bdd7a53892d507e1e92129c8f191de49ee8566f7d | status=RESOLVED | reason=None

Dependency graph



Dependency graph edge list

LM:S:ST2:location:20-40 -> RL:S:ST2:location:00:surface-landmark [executable_to_relation]
RL:S:ST2:location:00:surface-landmark -> S:ST2:location [relation_to_statement]
LM:S:ST2:location:7-11 -> RL:S:ST2:location:01:surface-landmark [executable_to_relation]
RL:S:ST2:location:01:surface-landmark -> S:ST2:location [relation_to_statement]
S:ST2:location -> ST2 [statement_to_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature
- [] distinct subfeatures
- [] relation operator
- [] condition state
- [] RESOLVED/UNRESOLVED
- [] ACCEPT
- [] REJECT

BL40: RESOLVED -> RESOLVED

Defect family: relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete

WHO source

location | S:BL40:location | page 128 | On the posterior aspect of the knee, at the midpoint of the popliteal crease.

Semantic decomposition

LM:S:BL40:location:56-76 | source=the popliteal crease | class=crease.skin_crease | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"1dbd d6b2b621a4ee0185cf93253f7062de8485746c285cadd13ae1177b60ff9","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:popliteal_crease","subfeature_type":null} | reason=None

LM:S:BL40:location:7-23 | source=posterior aspect | class=surface.aspect | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"12ffe e9bff3884af77a892f4d06484125686f5b1fc7f181d5dc0a994051fc63e","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:posterior_knee","subfeature_type":null} | reason=None

Composite binding view

Relation execution

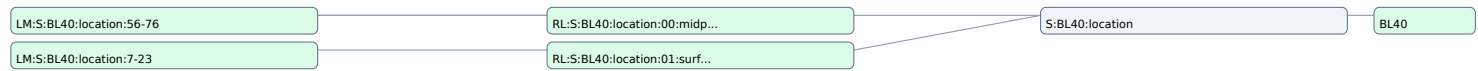
RL:S:BL40:location:00:midpoint-of-entity | op=entity_midpoint | semantic=['LM:S:BL40:location:56-76'] | executable=['LM:S:BL40:location:56-76'] | dir=None | branch=None | source=S:BL40:location | semantic_hash=040b844348afc2095a77e08a5665ab297e48ba3c231c87622966125fa4e6b524 | status=RESOLVED | reason=None

RL:S:BL40:location:01:surface-landmark | op=surface_membership | semantic=['LM:S:BL40:location:7-23'] | executable=['LM:S:BL40:location:7-23'] | dir=None | branch=None | source=S:BL40:location | semantic_hash=a21e46b9233fb5660cec96c1e628422d944aab136058e04b043f442b8f63e5e7 | status=RESOLVED | reason=None

Derived geometry outputs

{"derived_geometry_id":"DERIVED:RL:S:BL40:location:00:midpoint-of-entity:ebcdd629f27ccdb4","geometry_hash":"ebcdd629f27ccdb4172d4b83ee5b603b6d1 5456faa5fde5a39699c96983e39cc","operator":"entity_midpoint","producer_relation_id":"RL:S:BL40:location:00:midpoint-of-entity","semantic_argument_node_ids":["LM:S:BL40:location:56-76"],"source_statement_id":"S:BL40:location"}

Dependency graph



Dependency graph edge list

- LM:S:BL40:location:56-76 -> RL:S:BL40:location:00:midpoint-of-entity [executable_to_relation]
- RL:S:BL40:location:00:midpoint-of-entity -> S:BL40:location [relation_to_statement]
- LM:S:BL40:location:7-23 -> RL:S:BL40:location:01:surface-landmark [executable_to_relation]
- RL:S:BL40:location:01:surface-landmark -> S:BL40:location [relation_to_statement]
- LM:S:BL40:location:56-76 -> RL:S:BL40:location:00:midpoint-of-entity [operand_to_derived_operator]
- RL:S:BL40:location:00:midpoint-of-entity -> DERIVED:RL:S:BL40:location:00:midpoint-of-entity:ebcdd629f27ccdb4 [operator_to_derived_geometry]
- S:BL40:location -> BL40 [statement_to_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [] WHO source phrase
- [] semantic granularity
- [] child subfeature
- [] execution
- [] distinct subfeatures
- [] distinct executable identity
- [] relation operator
- [] operand
- [] condition state
- [] RESOLVED/UNRESOLVED
- [] ACCEPT
- [] REJECT

ST4: RESOLVED -> UNRESOLVED

Defect family: parent_only_composite_resolution, parent_only_composite_resolution, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, direction_preservation_failure, quantitative_relation_semantics_loss, quantitative_relation_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, parent_only_composite_resolution, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, direction_preservation_failure, relation_semantic_hash_mismatch, operand_trace_incomplete, parent_only_composite_resolution, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_semantic_hash_mismatch, condition_linkage_loss

WHO source

location | S:ST4:location | page 56 | On the face, 0.4 f-cun lateral to the angle of the mouth.

note | S:ST4:note:1 | page 56 | Note: Lateral to the angle of the mouth, the point is located in the nasolabial sulcus or on the continuation of the nasolabial sulcus.

Semantic decomposition

LM:S:ST4:location:33-56 | source= the angle of the mouth | class=orifice_or_cavity | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:ST4:location:7-11 | source=face | class=region.body_region | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA24728","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:ST4:note:1:16-39 | source= the angle of the mouth | class=orifice_or_cavity | parent=None | child=None | executor=bound_composite_subfeature | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:ST4:note:1:69-86 | source=nasolabial sulcus | class=soft_tissue_feature | parent=None | child=None | executor=approved_surface_registry | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=No approved BodyParts3D geometric definition; deliberately unresolved.

LM:S:ST4:note:1:97-134 | source=continuation of the nasolabial sulcus | class=soft_tissue_feature | parent=None | child=None | executor=approved_surface_registry | status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=No approved BodyParts3D geometric definition; deliberately unresolved.

Composite binding view

Relation execution

RL:S:ST4:location:00:relative-to | op=directional_relation | semantic=['LM:S:ST4:location:33-56'] | executable=[] | dir=lateral | branch=None | source=S:ST4:location | semantic_hash=fecd99e952ae5cfd7643d980abc18ed935b3c97fdf3a673e2d380e2b62f5bb52 | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:ST4:location:01:surface-landmark | op=surface_membership | semantic=['LM:S:ST4:location:7-11'] | executable=['LM:S:ST4:location:7-11'] | dir=None | branch=None | source=S:ST4:location | semantic_hash=b6ba39eef63501551822036a7ee6f2be3f05c66264e84b729f2ecf7d1a7cb3de | status=RESOLVED | reason=None

RL:S:ST4:note:1:00:relative-to | op=directional_relation | semantic=['LM:S:ST4:note:1:16-39'] | executable=[] | dir=lateral | branch=None | source=S:ST4:note:1 | semantic_hash=1191b9fc177ee29f1073bfda2248df9285c8fc092d51c9589912bae3deaa1bd1 | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:ST4:note:1:01:surface-landmark | op=surface_membership | semantic=['LM:S:ST4:note:1:69-86'] | executable=['LM:S:ST4:note:1:69-86'] | dir=None | branch=None | source=S:ST4:note:1 | semantic_hash=3ecacabb0eeb1529c9ebf5e4babe7f22dd8de009086b32aee5ee4df980ac4e0f | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:ST4:note:1:02:surface-landmark | op=surface_membership | semantic=['LM:S:ST4:note:1:97-134'] | executable=['LM:S:ST4:note:1:97-134'] | dir=None | branch=None | source=S:ST4:note:1 | semantic_hash=9e03a561e789bbee4a7aff60751353c229f9dfae21e61f98cf63e63f5d950564 | status=UNRESOLVED | reason=executable operand not uniquely executable

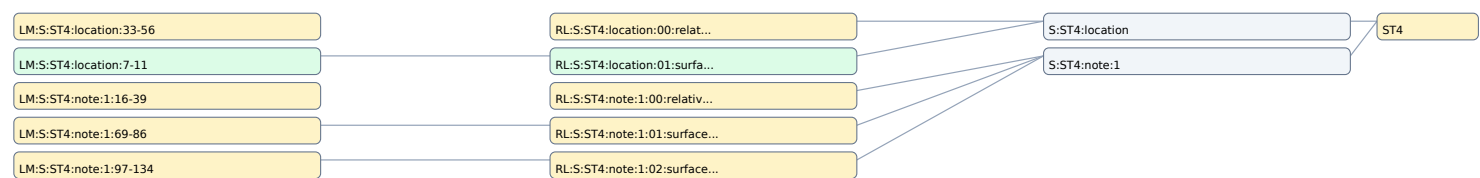
Quantitative measurement semantics

{ "anchor_landmark_id": "LM:S:ST4:location:33-56", "direction": "lateral", "executor": "approved_calibration", "measurement_id": "PM:S:ST4:location:00", "reason": null, "source_statement_id": "S:ST4:location", "source_unit": "F-cun", "status": "RESOLVED", "unit": "F-cun", "value": 0.4, "who_pdf_page": 56 }

Conditions

{ "branch_id": "branch_1", "condition_id": "CD:S:ST4:note:1:00", "condition_type": "alternative", "default_pose_compatible": null, "reason": "condition preserved as non-executed conditional branch", "repaired_status": "CONDITIONAL", "source_raw": "or on the continuation", "source_statement_id": "S:ST4:note:1" }

Dependency graph



Dependency graph edge list

PM:S:ST4:location:00 -> RL:S:ST4:location:00:relative-to [measurement_to_relation]
RL:S:ST4:location:00:relative-to -> S:ST4:location [relation_to_statement]
LM:S:ST4:location:7-11 -> RL:S:ST4:location:01:surface-landmark [executable_to_relation]
RL:S:ST4:location:01:surface-landmark -> S:ST4:location [relation_to_statement]
RL:S:ST4:note:1:00:relative-to -> S:ST4:note:1 [relation_to_statement]
LM:S:ST4:note:1:69-86 -> RL:S:ST4:note:1:01:surface-landmark [executable_to_relation]
RL:S:ST4:note:1:01:surface-landmark -> S:ST4:note:1 [relation_to_statement]
LM:S:ST4:note:1:97-134 -> RL:S:ST4:note:1:02:surface-landmark [executable_to_relation]
RL:S:ST4:note:1:02:surface-landmark -> S:ST4:note:1 [relation_to_statement]
LM:S:ST4:location:33-56 -> PM:S:ST4:location:00 [anchor_to_measurement]
PM:S:ST4:location:00 -> S:ST4:location [measurement_to_statement]
S:ST4:location -> ST4 [statement_to_point]
S:ST4:note:1 -> ST4 [statement_to_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

[] WHO source phrase
[] semantic granularity
[] child subfeature
[] distinct subfeatures
[] distinct executable identity
[] relation operator
[] condition state
[] RESOLVED/UNRESOLVED
[] ACCEPT
[] REJECT

TE6: RESOLVED -> RESOLVED

Defect family: relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, direction_preservation_failure, quantitative_relation_semantics_loss, quantitative_relation_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, relation_nonoperand_semantics_loss, quantitative_relation_semantics_loss, relation_semantic_hash_mismatch, operand_trace_incomplete

WHO source

location | S:TE6:location | page 169 | On the posterior aspect of the forearm, midpoint of the interosseous space between the radius and the ulna, 3 B-cun proximal to the dorsal wrist crease.

note | S:TE6:note:1 | page 169 | Note: 1 B-cun proximal to TE5, between the radius and the ulna, at the same level as TE7.

Semantic decomposition

LM:S:TE6:location:127-151 | source= the dorsal wrist crease | class=crease.skin_crease | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"0ff534739841ae12b463e9ea5102d7801e5de256871212f7ddeb84a0b0265288","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:dorsal_wrist_crease","subfeature_type":null} | reason=None

LM:S:TE6:location:52-106 | source=the interosseous space between the radius and the ulna | class=space.anatomical_space | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"4b89dd3e588ceb0c6fc9aae7de7f8c71ea842a9e0305158cad97d76f12f35f33","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:radius_ulna_interosseous_space","subfeature_type":null} | reason=None

LM:S:TE6:location:7-23 | source=posterior aspect | class=surface.aspect | parent=None | child=None | executor=approved_surface_registry | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":"7809ff04bb161d448c5b90dbbb407ddc76c10e74e85ee7fe0c795e866ebad73c","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"SR:posterior_forearm","subfeature_type":null} | reason=None

LM:S:TE6:location:87-93 | source=radius | class=bone.bone | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA23463","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:TE6:location:98-106 | source=the ulna | class=bone.bone | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA23466","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:TE6:note:1:25-29 | source= TE5 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":"TE5","subfeature_type":null} | reason=None

LM:S:TE6:note:1:43-49 | source=radius | class=bone.bone | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA23463","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:TE6:note:1:54-62 | source=the ulna | class=bone.bone | parent=None | child=None | executor=fma_mesh | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":"FMA23466","geometry_hash":null,"kind":"fma_concept","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

LM:S:TE6:note:1:85-88 | source=tE7 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"derived_geometry_id":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":null,"part_id":null,"producer_relation_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=None

Composite binding view

Relation execution

RL:S:TE6:location:02:midpoint-of-entity | op=entity_midpoint | semantic=['LM:S:TE6:location:52-106'] | executable=['LM:S:TE6:location:52-106'] | dir=None | branch=None | source=S:TE6:location | semantic_hash=d48745ca38495e5195e398747d052d4b434517184a88e9cd53e3706c5ab4dd94 | status=RESOLVED | reason=None

RL:S:TE6:location:00:between | op=between_constraint | semantic=['LM:S:TE6:location:87-93', 'LM:S:TE6:location:98-106'] | executable=['LM:S:TE6:location:87-93', 'LM:S:TE6:location:98-106'] | dir=None | branch=None | source=S:TE6:location | semantic_hash=f4bb89b887304a01961f487c126eff873313109a8eb93156d9b6171110d02fbc | status=RESOLVED | reason=None

RL:S:TE6:location:01:relative-to | op=directional_relation | semantic=['LM:S:TE6:location:127-151'] | executable=['LM:S:TE6:location:127-151'] | dir=proximal | branch=None | source=S:TE6:location | semantic_hash=4bc4faeb0b91e9df11acadec51991c4357cc541dee9ed68372f820a30056015d | status=RESOLVED | reason=None

RL:S:TE6:location:03:surface-landmark | op=surface_membership | semantic=['LM:S:TE6:location:7-23'] | executable=['LM:S:TE6:location:7-23'] | dir=None | branch=None | source=S:TE6:location | semantic_hash=023e5b5d7d041639a6c8dd43496a402ec674d9bb1939206410e8da6ff08c9f9a | status=RESOLVED | reason=None

RL:S:TE6:note:1:00:between | op=between_constraint | semantic=['LM:S:TE6:note:1:43-49', 'LM:S:TE6:note:1:54-62'] | executable=['LM:S:TE6:note:1:43-49', 'LM:S:TE6:note:1:54-62'] | dir=None | branch=None | source=S:TE6:note:1 | semantic_hash=9ff8843b3054442b9da374560ec65931382902ac1caaa1d7b5f5602fa1724021 | status=RESOLVED | reason=None

RL:S:TE6:note:1:01:relative-to | op=directional_relation | semantic=['LM:S:TE6:note:1:25-29'] | executable=['LM:S:TE6:note:1:25-29'] | dir=proximal | branch=None | source=S:TE6:note:1 | semantic_hash=a220920b811d82795f9fe01b7b88b77c35b29c51190a5cd176e292461e9fc157 | status=RESOLVED | reason=None

RL:S:TE6:note:1:02:same-level | op=same_level_plane | semantic=['LM:S:TE6:note:1:85-88'] | executable=['LM:S:TE6:note:1:85-88'] | dir=None | branch=None | source=S:TE6:note:1 | semantic_hash=71614e8ea7bacd1a311d98193765d53251940bf62a7afe28e1f45cd1e2060b1a | status=RESOLVED | reason=None

RL21:S:TE6:note:1:26-29:reference-acupoint:87571bdea7 | op=reference_dependency | semantic=['LM:S:TE6:note:1:25-29'] | executable=['LM:S:TE6:note:1:25-29'] | dir=None | branch=None | source=S:TE6:note:1 | semantic_hash=4f9b6e88b9fa76124407450a1b4a2a6916ed2df2793a2fd6f6765446eb1bddaf | status=RESOLVED | reason=None

RL21:S:TE6:note:1:85-88:reference-acupoint:eeb3b40f1e | op=reference_dependency | semantic=['LM:S:TE6:note:1:85-88'] | executable=['LM:S:TE6:note:1:85-88'] | dir=None | branch=None | source=S:TE6:note:1 | semantic_hash=c90d63a56fe762d3e2f939e04e7337da78727da2f7e23bb517fa0d08bf9de1d9 | status=RESOLVED | reason=None

Quantitative measurement semantics

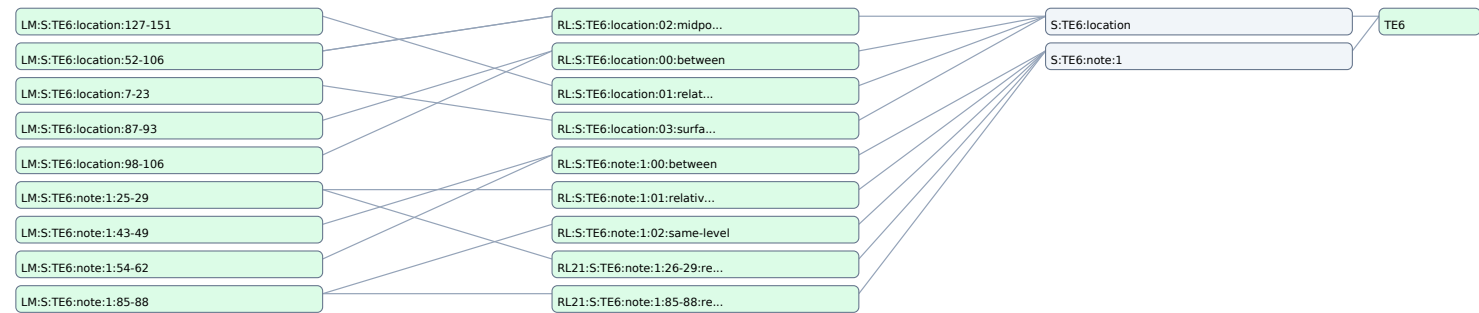
{ "anchor_landmark_id": "LM:S:TE6:location:127-151", "direction": "proximal", "executor": "approved_calibration", "measurement_id": "PM:S:TE6:location:00", "reason": null, "source_statement_id": "S:TE6:location", "source_unit": "B-cun", "status": "RESOLVED", "unit": "B-cun", "value": 3.0, "who_pdf_page": 169 }

{ "anchor_landmark_id": "LM:S:TE6:note:1:25-29", "direction": "proximal", "executor": "approved_calibration", "measurement_id": "PM:S:TE6:note:1:00", "reason": null, "source_statement_id": "S:TE6:note:1", "source_unit": "B-cun", "status": "RESOLVED", "unit": "B-cun", "value": 1.0, "who_pdf_page": 169 }

Derived geometry outputs

{ "derived_geometry_id": "DERIVED:RL:S:TE6:location:02:midpoint-of-entity:c96d933a14827638", "geometry_hash": "c96d933a14827638443b0e185fde00808f7be061673e58d72834ac8454890f2f", "operator": "entity_midpoint", "producer_relation_id": "RL:S:TE6:location:02:midpoint-of-entity", "semantic_argument_node_ids": ["LM:S:TE6:location:52-106"], "source_statement_id": "S:TE6:location" }

Dependency graph



Dependency graph edge list

LM:S:TE6:location:52-106 -> RL:S:TE6:location:02:midpoint-of-entity [executable_to_relation]
RL:S:TE6:location:02:midpoint-of-entity -> S:TE6:location [relation_to_statement]
LM:S:TE6:location:87-93 -> RL:S:TE6:location:00:between [executable_to_relation]
LM:S:TE6:location:98-106 -> RL:S:TE6:location:00:between [executable_to_relation]
RL:S:TE6:location:00:between -> S:TE6:location [relation_to_statement]
LM:S:TE6:location:127-151 -> RL:S:TE6:location:01:relative-to [executable_to_relation]
PM:S:TE6:location:00 -> RL:S:TE6:location:01:relative-to [measurement_to_relation]
RL:S:TE6:location:01:relative-to -> S:TE6:location [relation_to_statement]
LM:S:TE6:location:7-23 -> RL:S:TE6:location:03:surface-landmark [executable_to_relation]
RL:S:TE6:location:03:surface-landmark -> S:TE6:location [relation_to_statement]
LM:S:TE6:note:1:43-49 -> RL:S:TE6:note:1:00:between [executable_to_relation]
LM:S:TE6:note:1:54-62 -> RL:S:TE6:note:1:00:between [executable_to_relation]
RL:S:TE6:note:1:00:between -> S:TE6:note:1 [relation_to_statement]
LM:S:TE6:note:1:25-29 -> RL:S:TE6:note:1:01:relative-to [executable_to_relation]
PM:S:TE6:note:1:00 -> RL:S:TE6:note:1:01:relative-to [measurement_to_relation]
RL:S:TE6:note:1:01:relative-to -> S:TE6:note:1 [relation_to_statement]
LM:S:TE6:note:1:85-88 -> RL:S:TE6:note:1:02:same-level [executable_to_relation]
RL:S:TE6:note:1:02:same-level -> S:TE6:note:1 [relation_to_statement]
LM:S:TE6:note:1:25-29 -> RL21:S:TE6:note:1:26-29:reference-acupoint:87571bdea7 [executable_to_relation]
RL21:S:TE6:note:1:26-29:reference-acupoint:87571bdea7 -> S:TE6:note:1 [relation_to_statement]

LM:S:TE6:note:1:85-88 -> RL21:S:TE6:note:1:85-88:reference-acupoint:eeb3b40f1e [executable_to_relation]
RL21:S:TE6:note:1:85-88:reference-acupoint:eeb3b40f1e -> S:TE6:note:1 [relation_to_statement]
LM:S:TE6:location:127-151 -> PM:S:TE6:location:00 [anchor_to_measurement]
PM:S:TE6:location:00 -> S:TE6:location [measurement_to_statement]
LM:S:TE6:note:1:25-29 -> PM:S:TE6:note:1:00 [anchor_to_measurement]
PM:S:TE6:note:1:00 -> S:TE6:note:1 [measurement_to_statement]
LM:S:TE6:location:52-106 -> RL:S:TE6:location:02:midpoint-of-entity [operand_to_derived_operator]
RL:S:TE6:location:02:midpoint-of-entity -> DERIVED:RL:S:TE6:location:02:midpoint-of-entity:c96d933a14827638 [operator_to_derived_geometry]
S:TE6:location -> TE6 [statement_to_point]
S:TE6:note:1 -> TE6 [statement_to_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

[] WHO source phrase
[] semantic granularity
[] child subfeature
[] child subfeature execution
[] distinct subfeatures
[] distinct subfeatures distinct executable identity
[] relation operator
[] relation operator operand
[] condition state
[] RESOLVED/UNRESOLVED
[] ACCEPT
[] REJECT